

PORTFOLIO

Faezeh Nikmandan

Junior architect | MSc in Architecture for Sustainability



I'm Faezeh Nikmandan, an architect with a strong interest in combining creative design with smart digital technologies. With a solid foundation in Building Information Modeling (BIM) for Architecture, I work with tools like Revit and AutoCAD to develop accurate, data-rich models and coordinated project documentation.

I see BIM as a powerful, collaborative process that enhances design precision, project efficiency, and sustainability. I'm dedicated to continuous learning and aim to contribute to forward-thinking, multidisciplinary teams with clarity, innovation, and purpose.

Faezeh Nikmandan

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faezeh-nikmandan



https://faezehnikmandan.github.io/



Education

- Politecnico di Torino** ,Torino, Italy
 - Master of Sciences in Architecture for sustainability 2021-2024
- Seraj University** , Tabriz ,Iran
 - Bachelor in Architecture Engineering 2015-2019

Work Experience

- Tangram Studio**
(Alice e Giorgio Giani Architetti) : Turin, Italy , On-site , Full-time
 - Junior Architect : Nov 2024 - Sep 2025
- Gas architettura** : Turin, Italy , On-site ,Part-time
 - Junior Architect : Aug 2023 - Mar 2024
 - Curriculum Internship : Jun 2023 - Aug 2023
- Scale Carpentry** : Iran , Hybrid , Full-time
 - Interior Designer : 2020 - 2021

Skills

- Language

 - English : Advanced
 - Italian : Elementary
 - Persian :Native
 - Turkish: Advanced
 - Azerbaijan : Native
- Software Proficiency

 - Revit : 8/10
 - Dynamo: 8/10
 - Rhino : 5/10
 - Grasshopper: 6/10
 - 3D Max : 8/10
 - AutoCad : 9/10
 - Camtasia : 9/10
 - Rendering
 - Lumion : 5/10
 - V-Ray : 7/10
 - Enscape : 8/10
 - Unreal Engine 6/10
- Adobe
 - Photoshop :9/10
 - InDesign: 9/10
 - Illustrator:9/10
 - Microsoft Office
 - Excel :7/10
 - PowerPoint : 9/10
 - Word:9/10
 - Programming Language
 - Python : 3/10
 - Html : 6/10
 - Css : 6/10
 - JavaScript : 3/10
 - React : 5/10

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Realizzazione di Nuovo Studentato

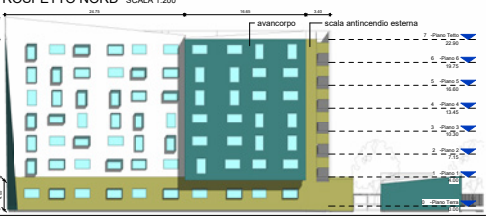
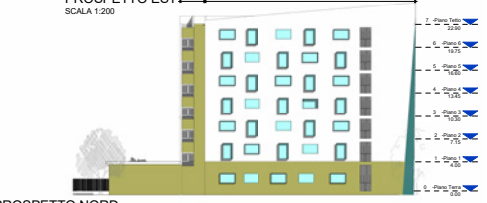
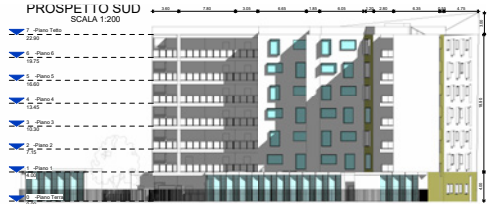
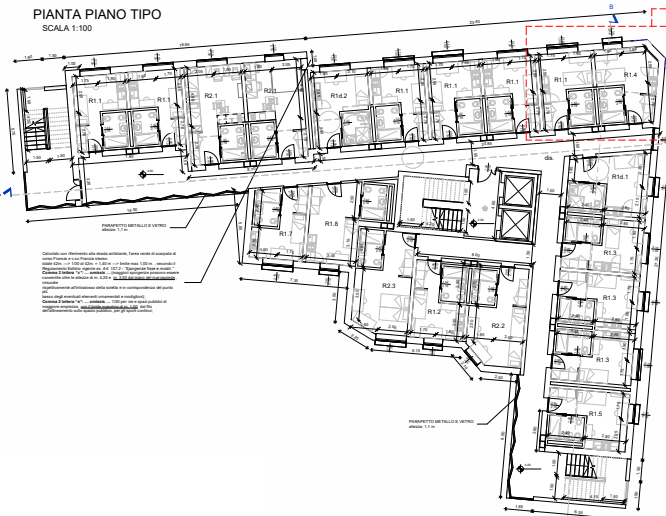
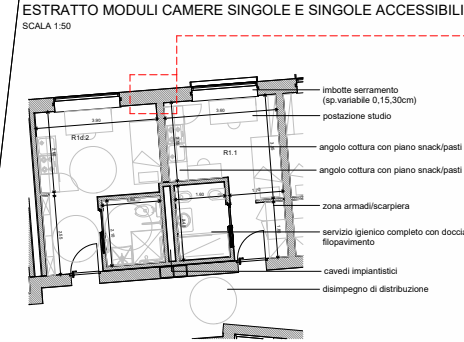
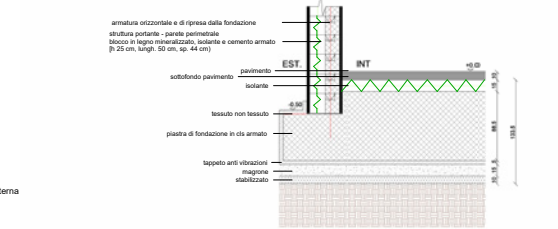
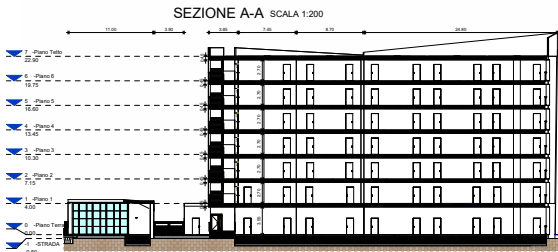
Year :2024-2025

Type : Professional Project

Location : Comune di collegno

The project involves the construction of a new student residence in Collegno (Turin), on Via Torino near Corso Francia. It will replace existing structures with a 7 story building offering student housing units along with common and service areas. The development is designed to provide modern accommodation for students in a well-connected but busy urban area.

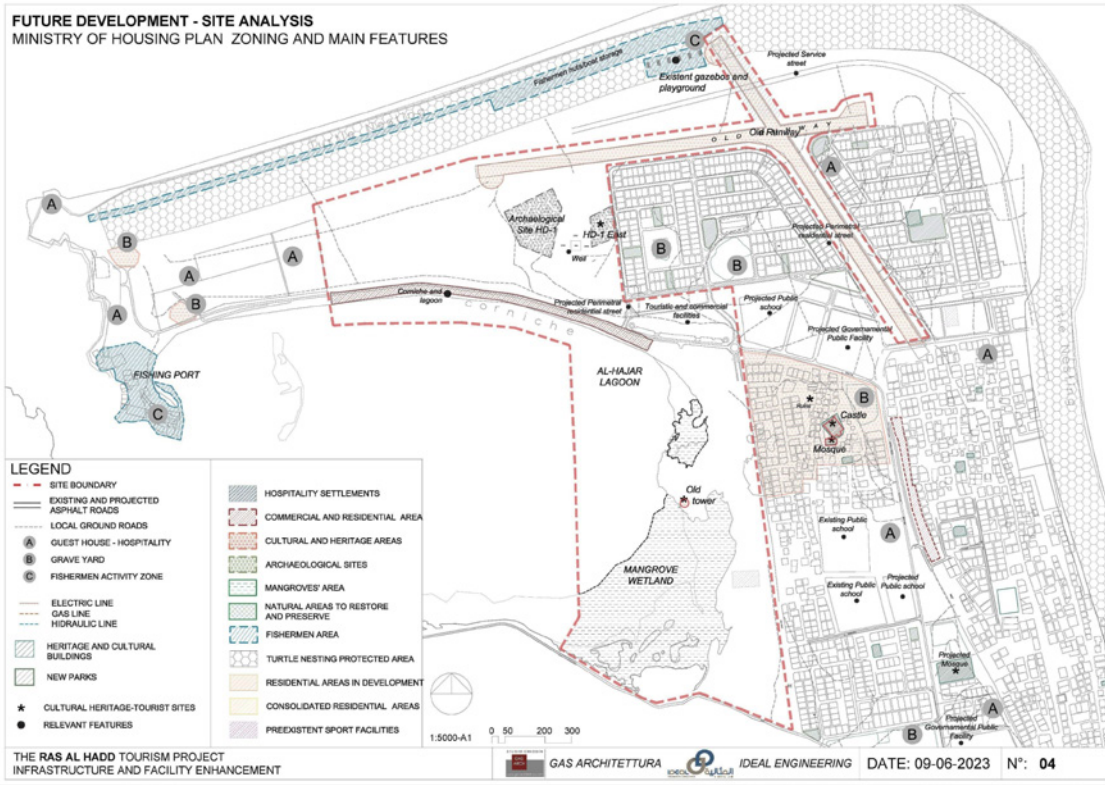
I contributed as an architect, working across different project phases. My responsibilities included creating detailed Revit models, coordinating existing conditions with design proposals, producing technical drawings (plans, sections, elevations), and generating computometric reports of building elements and spaces. I also developed 3D renderings and prepared professional presentations to communicate the design effectively.



Ras Al Hadd Improvement Project

Year :2023-2024
Type : Professional Project
Location : South Al Sharqiyah Governorate, Oman
Website : www.gasarchitettura.com

This project aimed to enhance tourism facilities in Ras Al Hadd to boost both domestic and international tourism through sustainable development. It focused on preserving the natural environment, including turtle nesting beaches, lagoon, and wetlands, while integrating cultural and historical assets like the fort, tower, and old runway. The project also sought to create opportunities for public and private investments and support local businesses through innovative improvements. By emphasizing ecological and cultural preservation, the project aimed to increase visitor numbers and create lasting positive impacts for the local community and the region's tourism sector.



As an architectural assistant at Gas Architettura Studio, I collaborated with the team on both the conceptual and detailed design stages, focusing on sustainable tourism infrastructure. Together, we conducted extensive research on the area's natural and cultural heritage, which informed our design decisions to balance development and conservation. Visual representations and presentations were created by me along with the team to effectively communicate design ideas to clients and stakeholders, making it easier for them to understand and approve the project plans. Additionally, detailed architectural drawings and documentation were prepared to ensure the project's execution and compliance with local regulations.



Club Med Sansicario Alto and Borgo

Year :2023-2024
Type : Professional Project
Location : Sansicario Torinese,Italy
Website: www.sunsicario.com , www.gasarchitettura.com

The project for the new Village Ski Hotel Resort CLUB MED in Sansicario is located in the municipality of Cesana Torinese, in the San Sicario Alto area, within the Via Lattea ski area. It includes a new hotel structure with 1,000 beds, a wellness center, a spa, swimming pools, and restaurants serving local specialties.



As an architectural assistant at Gas Architettura Studio, my responsibilities included:

- Creating 2D drawings, such as executive detailed plans, elevations and sections, using AutoCAD.
- Developing a detailed 3D model in Revit, including elevations, plans, and sections, as well as producing interior and exterior renders using Enscape.
- Producing isometric plans and incorporating proposed materials using Photoshop.



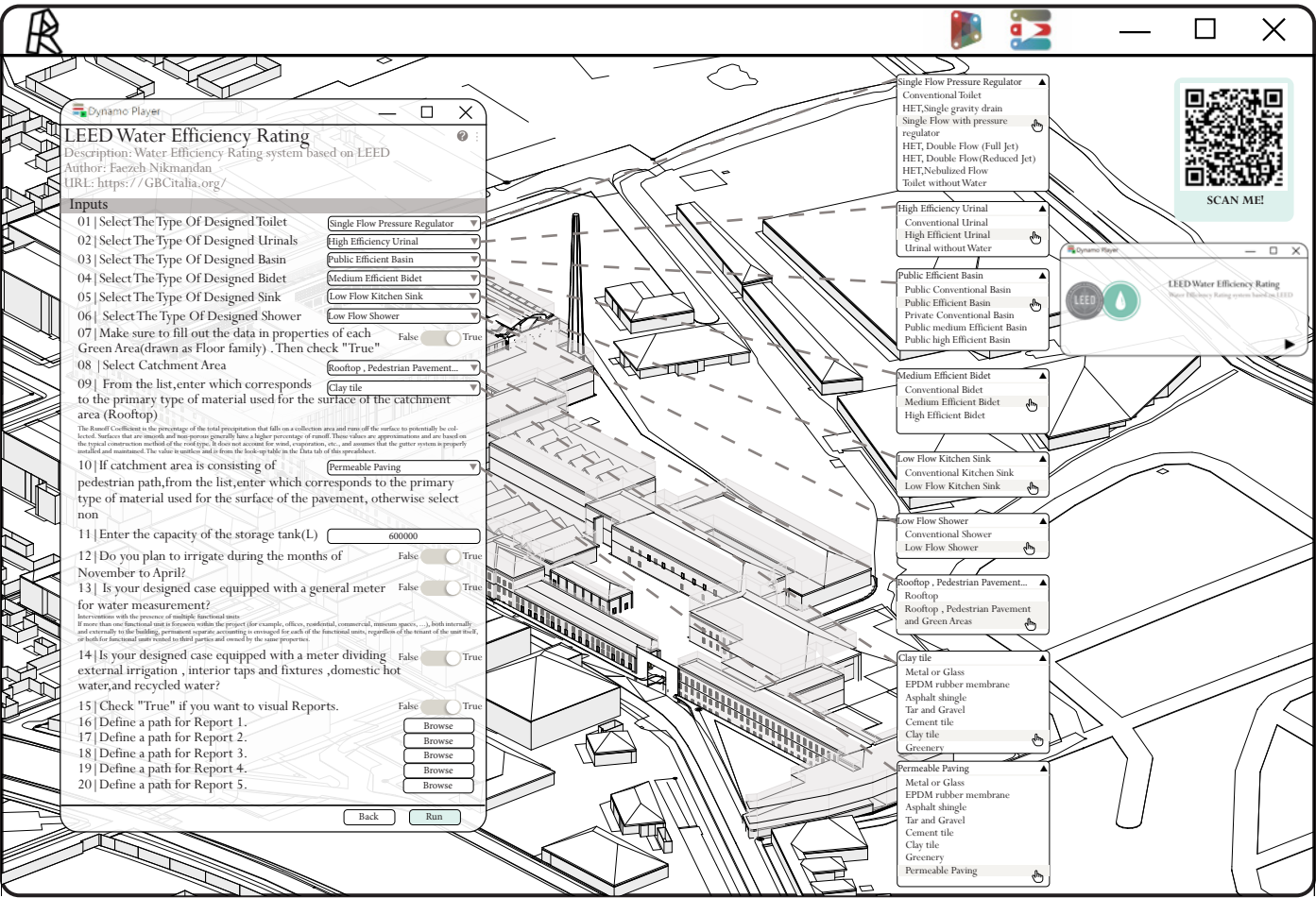
BIM and LEED Integration for Water Conservation

Year : 2024
Type : Master Thesis
Location :Turin ,Italy
Link : [Thesis](#)

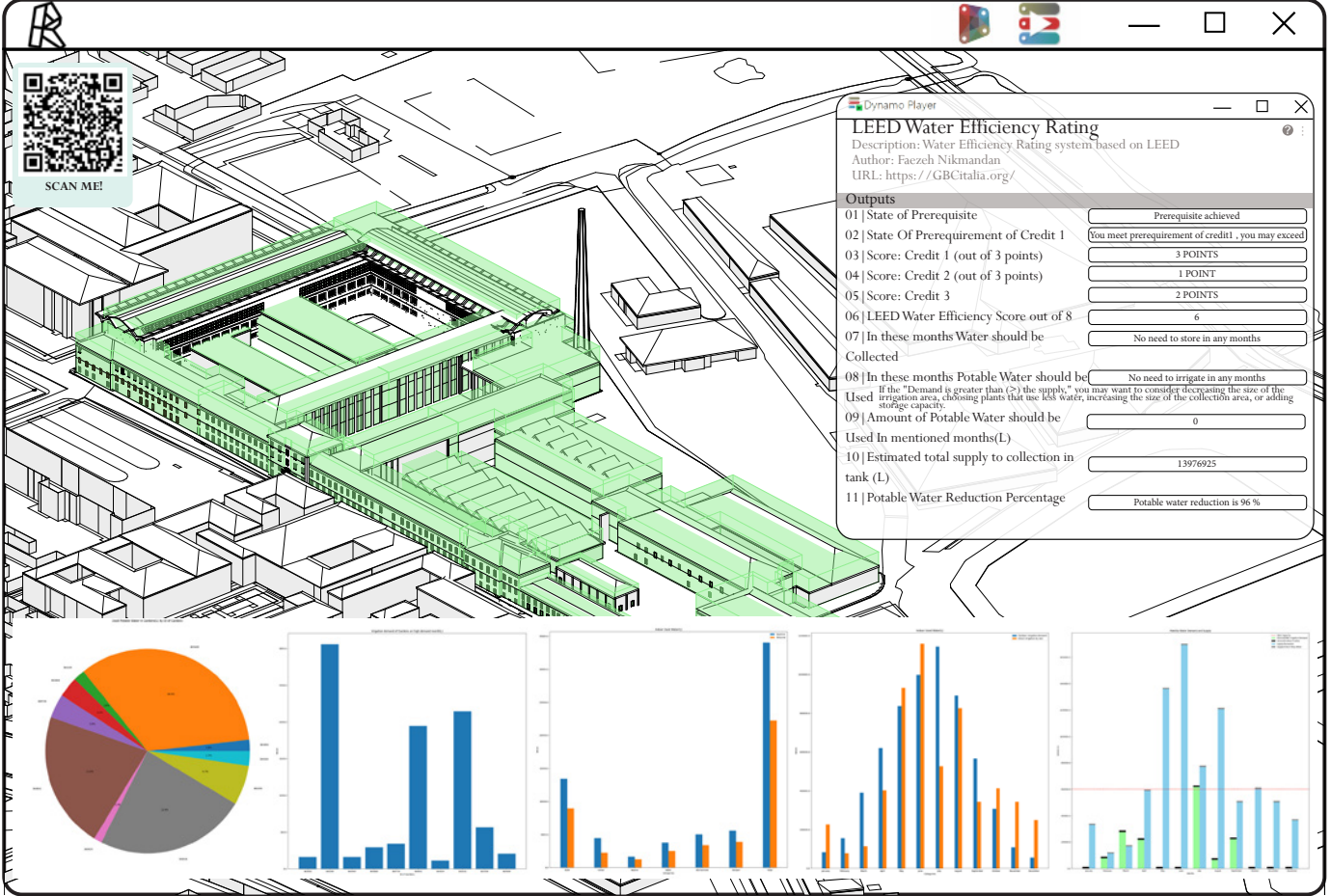
Professors :
Anna Osello
Matteo Del Giudice

The thesis examines the green building concept with a focus on water conservation, utilizing Building Information Modeling (BIM) to analyze this concept within the framework of sustainability by following the LEED certification system. The methodology, created in Dynamo,using python in some codes, for BIM modeling software, aims to simplify assessment parameters and calculations, assisting users in selecting appropriate strategies and parameters to acquire credits. This methodology was successfully implemented for Manifattura Tabacchi, facilitating the integration of potable water conservation within the LEED framework during the early stages of design.

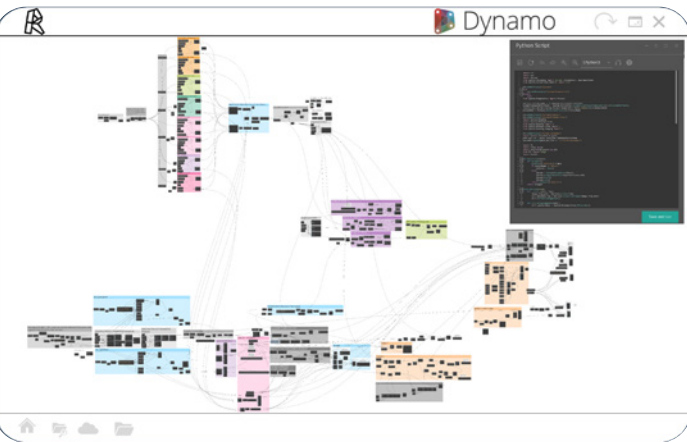
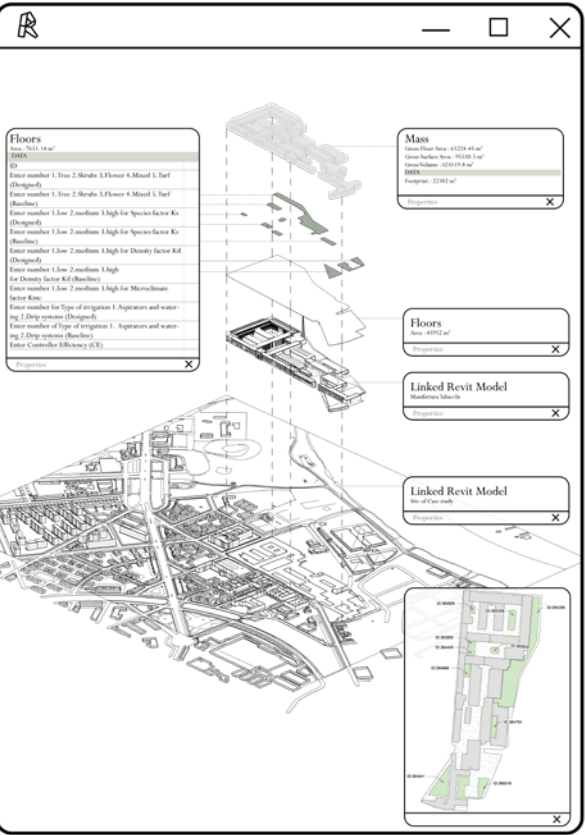
A conceptual model of a case study building was developed to be analyzed using various indoor water-consuming fixtures and outdoor landscaping choices.



The study calculated and analyzed both indoor and outdoor potable water usage, demonstrating that strategic design decisions can significantly conserve water. For validation, a comparative study was conducted across three scenarios, providing informative charts, monthly water consumption statistics, and reduction percentages. These scenarios confirmed the methodology's effectiveness using a low Level of Detail (LOD) model requirement, requiring minimal effort and easily obtainable inputs. It provides invaluable guidance to architects, landscape designers, owners, and other stakeholders, enabling them to make informed decisions during the preliminary and design phases of water-efficient projects and to determine the LEED score. A key aspect of the study is its emphasis on balancing water efficiency strategies with the fulfillment of LEED prerequisites.



It provides five informative charts and texts addressing critical questions about water usage, irrigation needs, storage adequacy, and LEED credit scores. Although this study focuses solely on the prerequisites and credits related to water efficiency, it suggests that automating the remaining LEED categories is viable. Achieving LEED certification requires a holistic approach that balances the water efficiency category with other LEED categories.



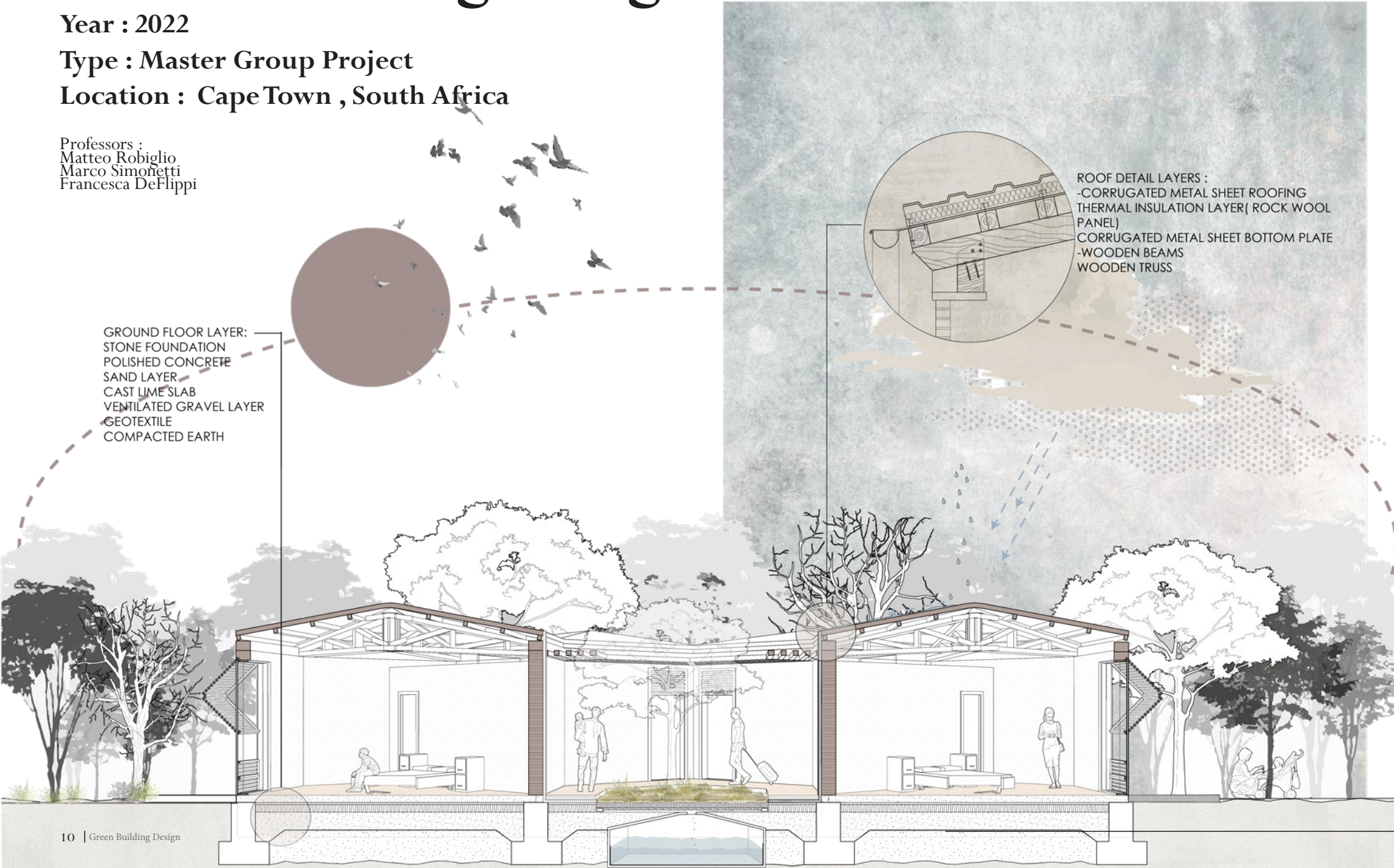
Green Building Design

Year : 2022

Type : Master Group Project

Location : Cape Town , South Africa

Professors :
Matteo Robiglio
Marco Simonetti
Francesca DeFlippi

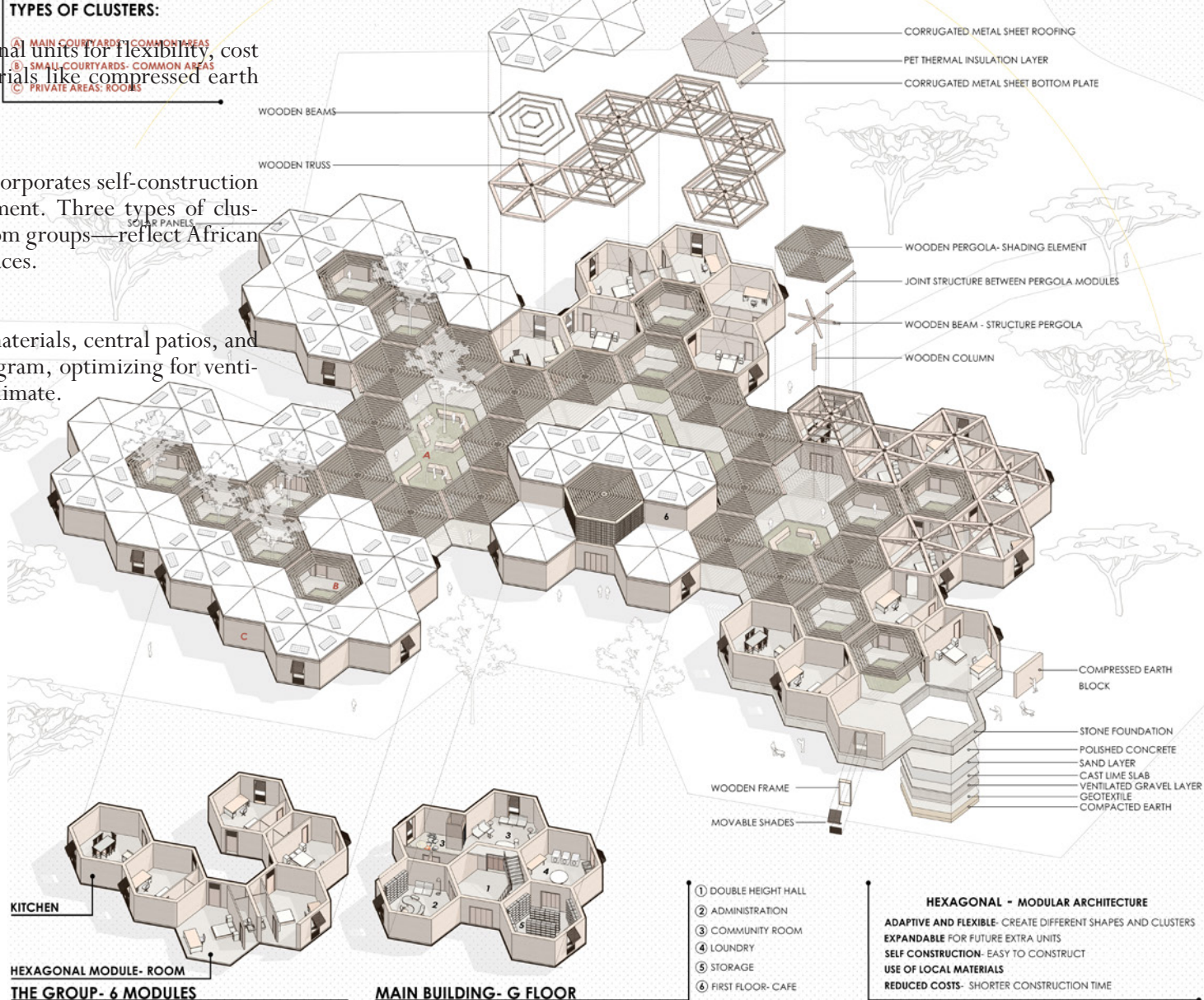


The Guest House project consists of six hexagonal clusters, each containing five rooms and a kitchen, all connected by a shaded wooden pergola and anchored by a central communal building.

The design emphasizes modular architecture with hexagonal units for flexibility, cost efficiency, and sustainability, using locally sourced materials like compressed earth blocks.

The hexagonal design supports future expansions and incorporates self-construction and participatory design, fostering community involvement. Three types of clusters—main courtyards, small courtyards, and private room groups—reflect African vernacular architecture's focus on communal outdoor spaces.

Bioclimatic strategies, including cross ventilation, local materials, central patios, and pergolas, were evaluated using the "Design Builder" program, optimizing for ventilation, shading, and material suitability to Cape Town's climate.



Restoration and Integrated Conservation

Year : 2022

Type : Master Group Project

Location : Turin ,Italy

Professors :
Antoni Teresa Spano
Carla Bartolozzi
Alessandro Pasquale Fantilli



Territorial Framework

Location

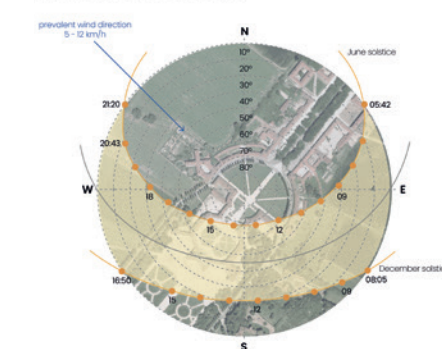


Green belt - Corona delle Delizie



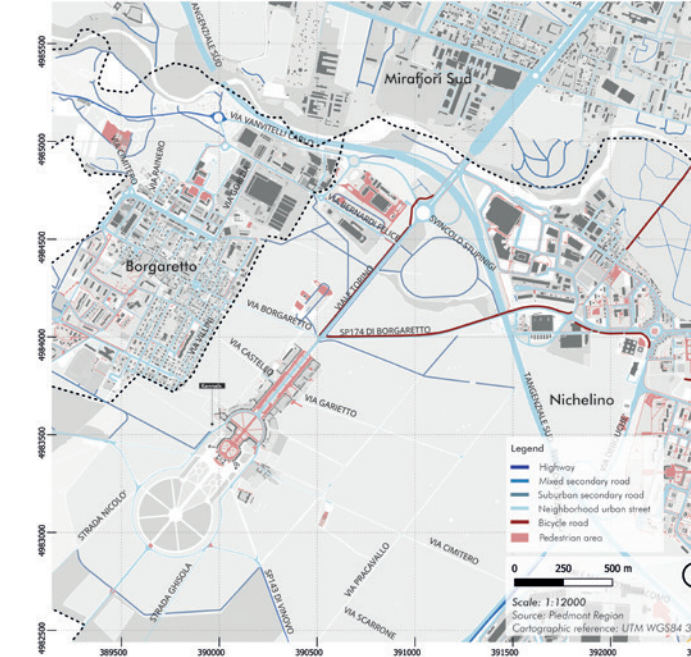
The "Corona delle delizie" is a set of royal residences that historically belonged to the Savoya family. Today, these residences are connected with the city's green infrastructure by means of a 90km bicycle route, which aims to rebuild the Turin metropolitan area and improve its quality of life. The "Green belt" route becomes a strategic project to attract tourists to the Hunting lodge.

Sunpath and wind analysis

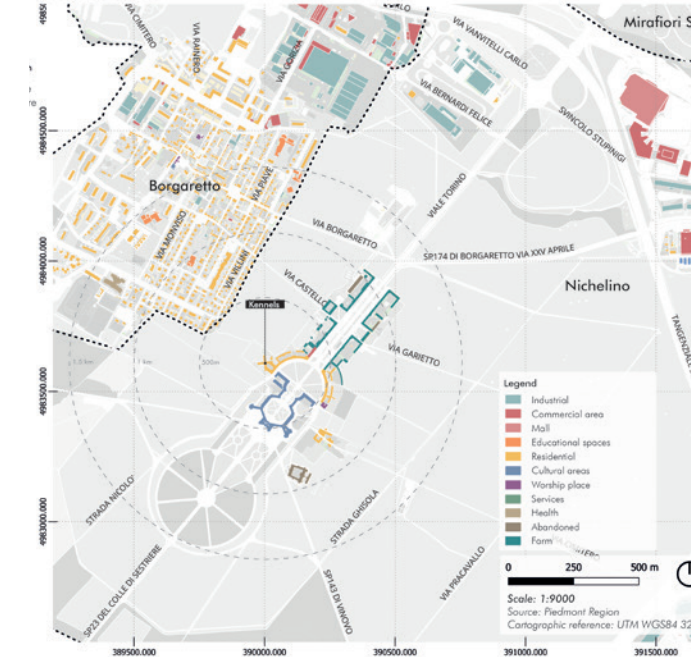


Taking into account that the physical degradation of materials can be caused by sunlight, wind and low temperatures, this diagram shows that the facades are more exposed to these conditions.

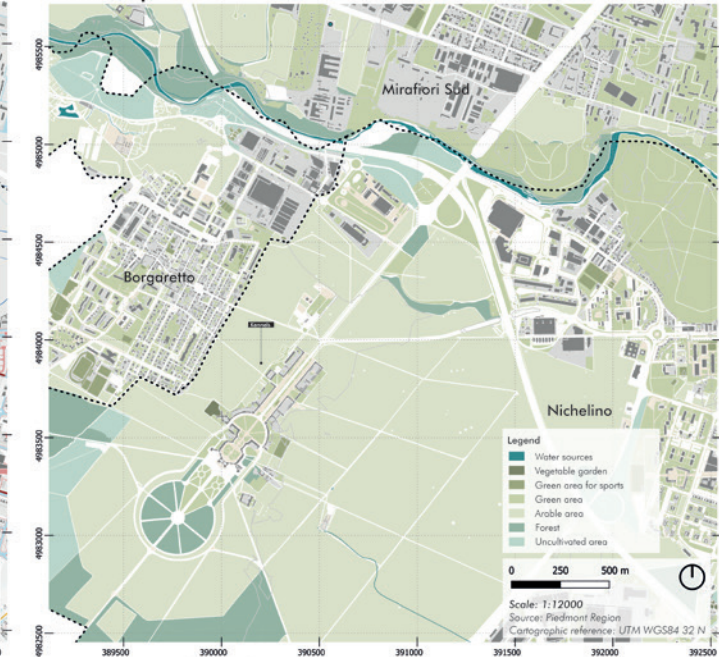
Mobility



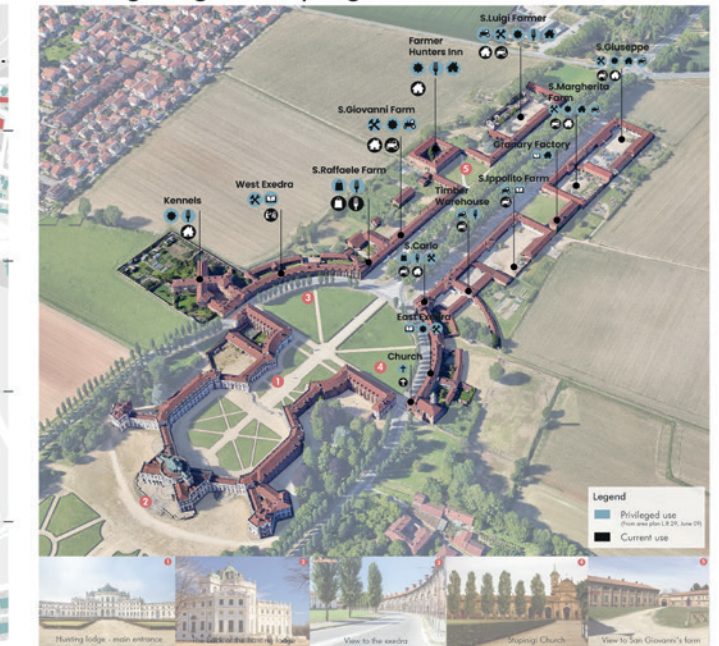
Land use



Green spaces



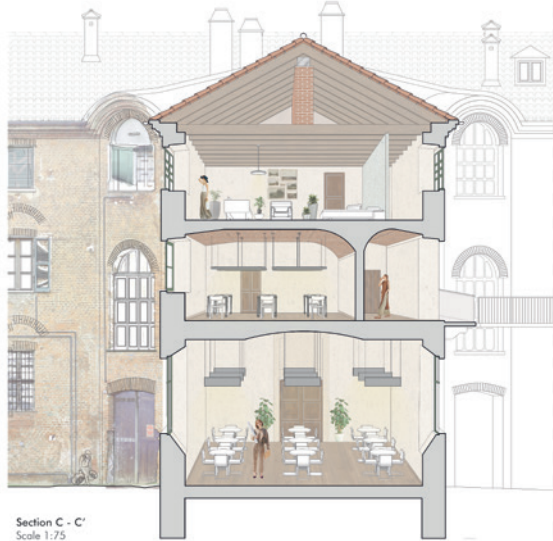
Hunting Lodge of Stupinigi



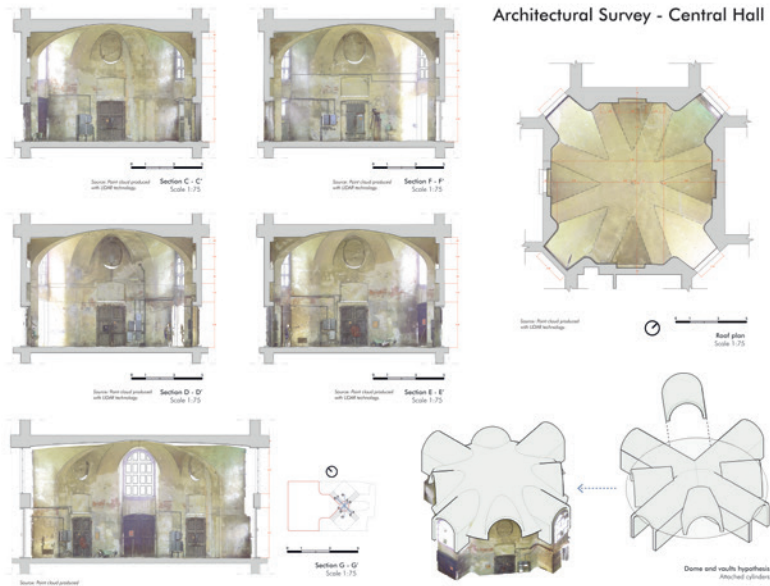
Architectural Design

Zero km - Kennels

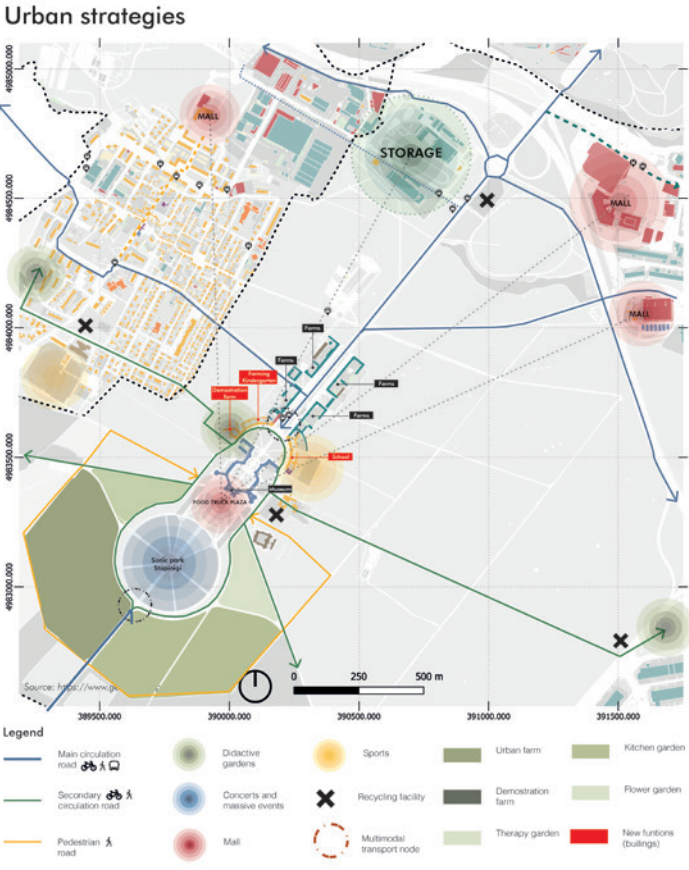
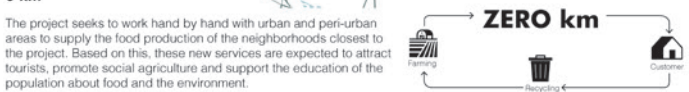
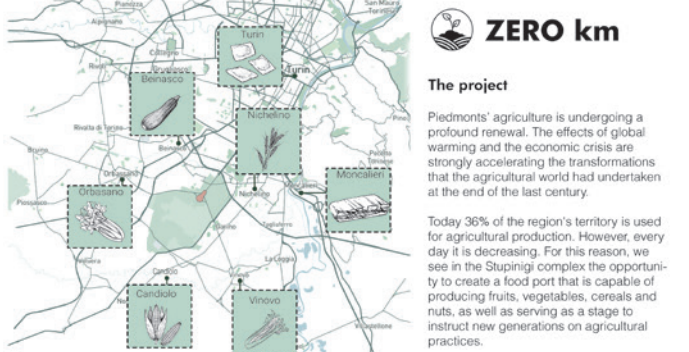
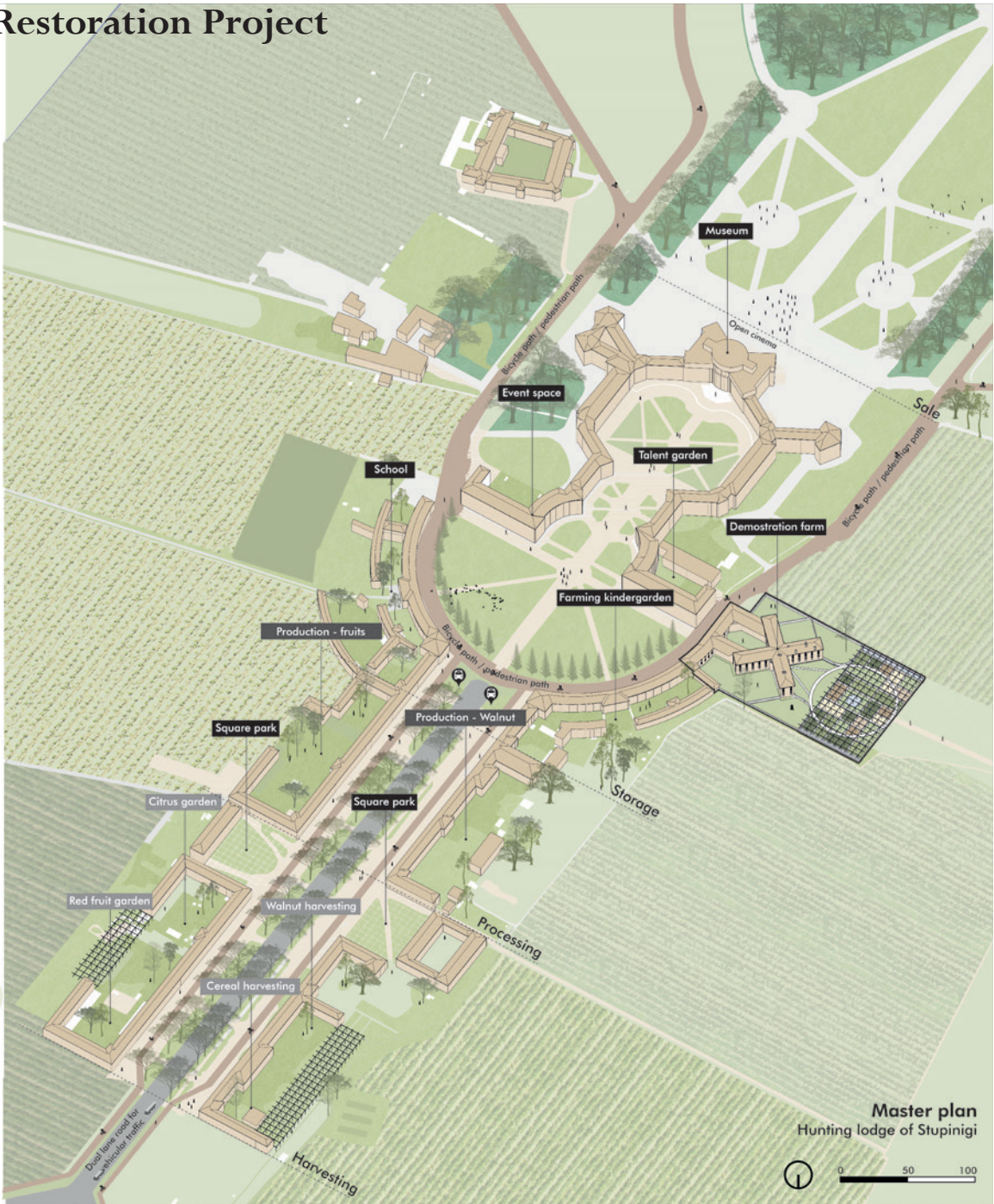
The project seeks to intervene the space minimally,since the gardens found in the backyard reflect the sense of belonging and community of its inhabitants. For this reason ,it is proposed to convert this space into an educational space to teach about the flora of the piedmont region.



Section B - B' Scale 1:200



Restoration Project



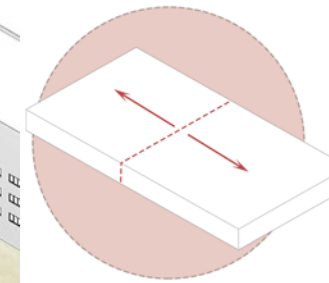
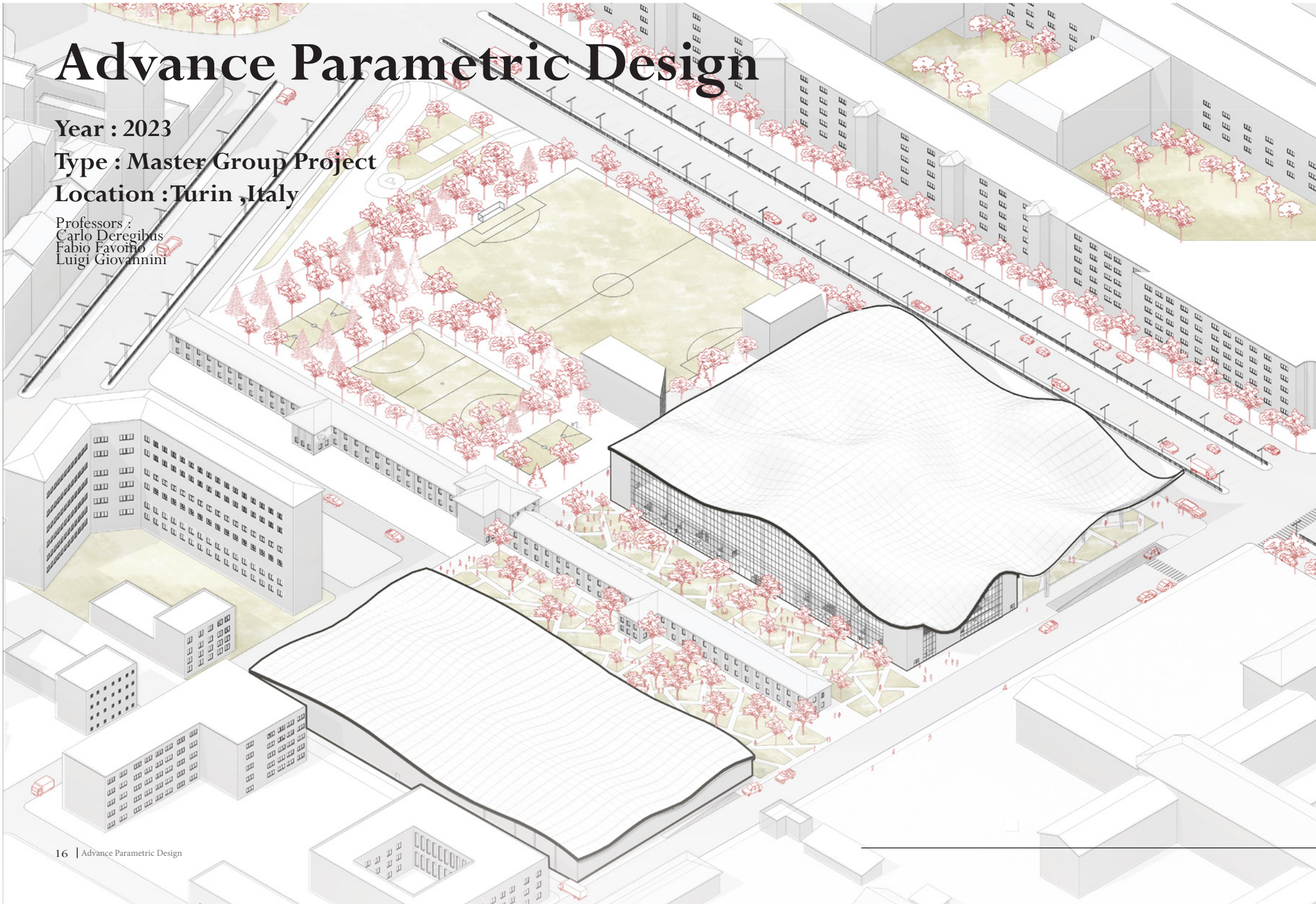
Advance Parametric Design

Year : 2023

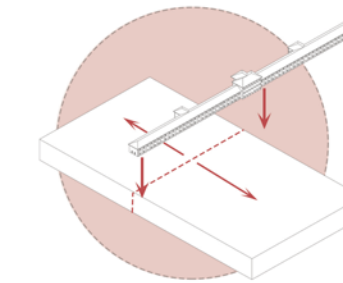
Type : Master Group Project

Location : Turin ,Italy

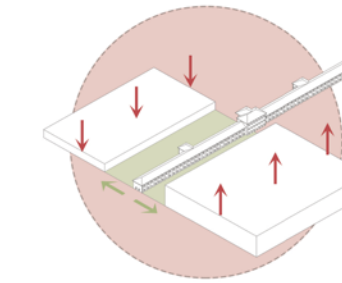
Professors :
Carlo Deregibus
Fabio Favotto
Luigi Giovannini



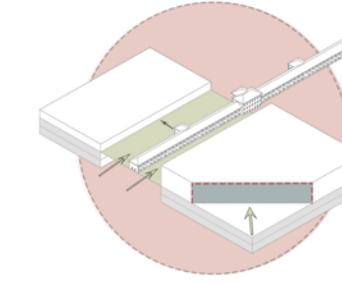
Simple Form with Divisions of Spaces And Areas as per requirements, i.e., The Convention Center and the Supermarket.



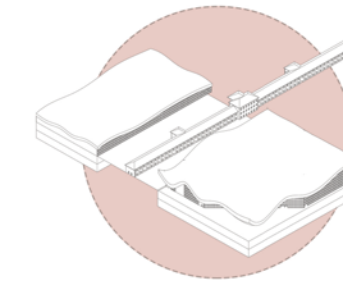
The Existing Historical Building (Barracks) to serve as a Transition space between the Convention Center and the Supermarket.



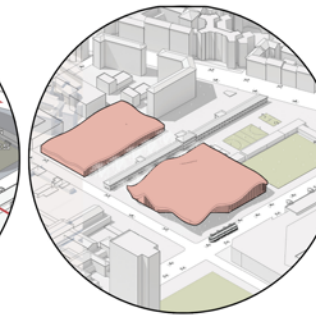
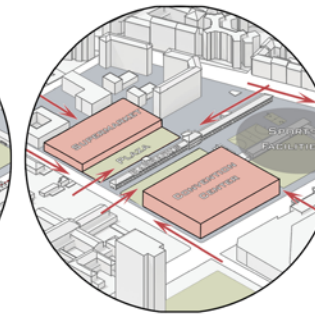
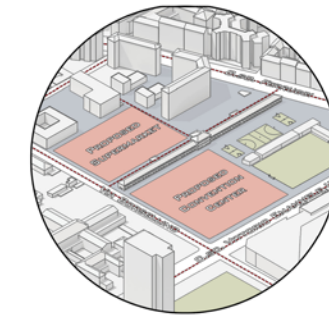
Alternating heights for the buildings and the Division by the Barracks make way for a Plaza on either sides of the buildings.



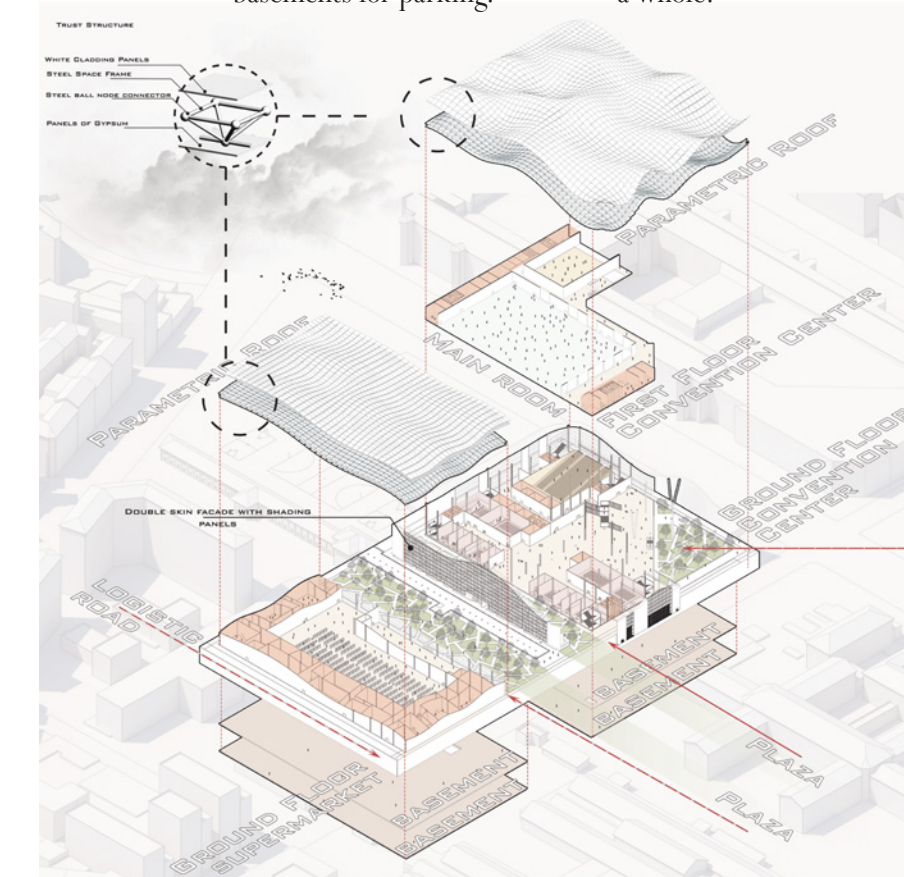
Angular access to the Convention Center and given entry-ways for easy access to the plaza and the buildings with the addition of basements for parking.

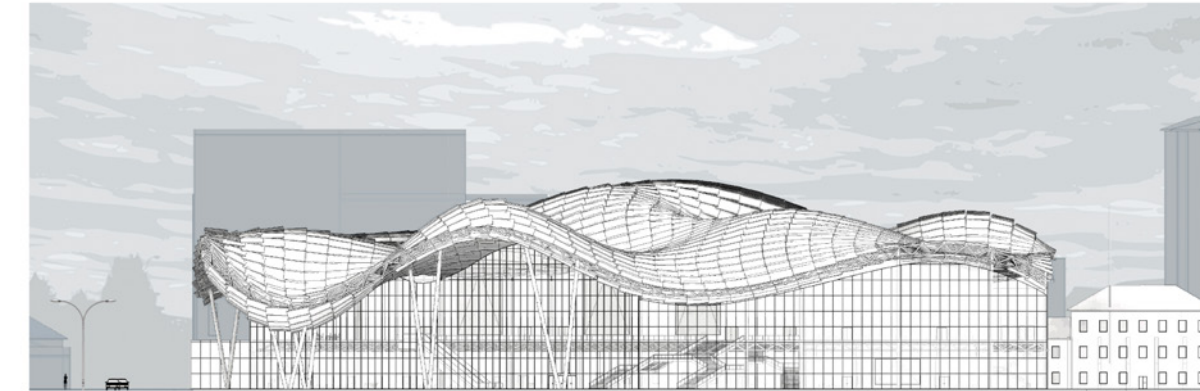
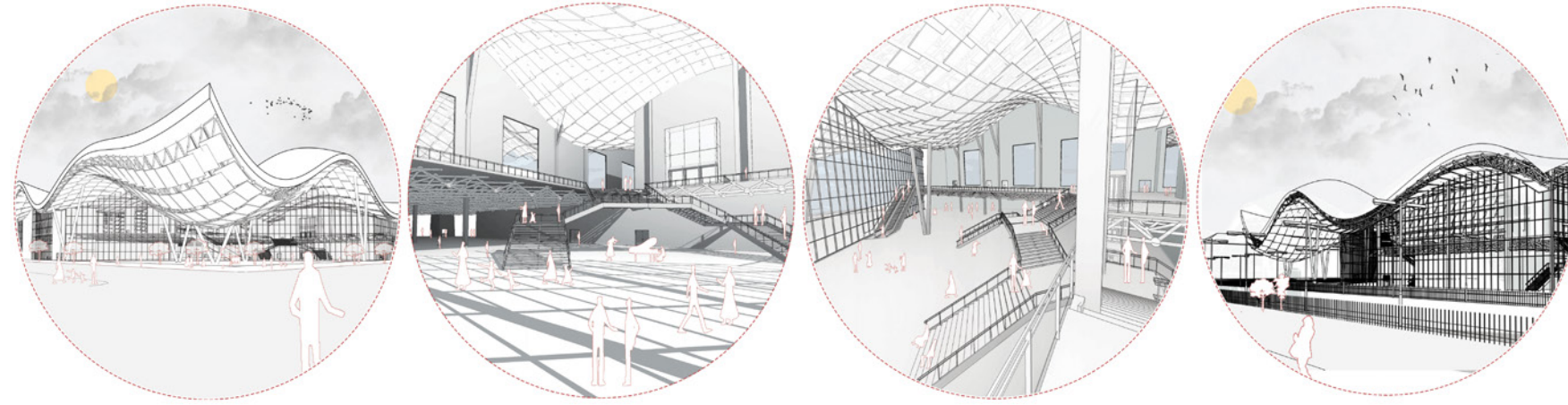


Curved Roof adding uniqueness and Uniformity between the two buildings And conveying a sense of simplicity for the project as a whole.



The project focuses on optimizing space allocation and access around the site, segregating spaces by function to enhance contextual presence, and ensuring a harmonious dialogue between built and green spaces. The design compensates for utilized green areas with minimal intervention and highlights a historical building as a buffer and plaza centerpiece to foster human interaction. The existing infrastructure is largely retained, and the design promotes unity and functionality between buildings. The plaza, serving as a car-free transitional zone between key structures, incorporates landscaping to connect with nearby green spaces like Giardino Nicola Groza and the Square at C.So. Ferrucci.





Specific Targets of My Daylighting Design:

- Ensure adequate daylight levels in a significant fraction of rooms for occupants to perform visual tasks.
- Utilize daylight to reduce energy use from electric lighting systems.
- Provide a comfortable and stimulating luminance distribution pattern, even in worst-case scenarios.
- Prevent glare in rooms.
- Avoid thermal discomfort from direct solar radiation on occupants.
- Prevent summer overheating to reduce energy use for cooling systems.
- Achieve high performance glass that reflects 47% of solar energy while allowing over 61% visible light transmittance

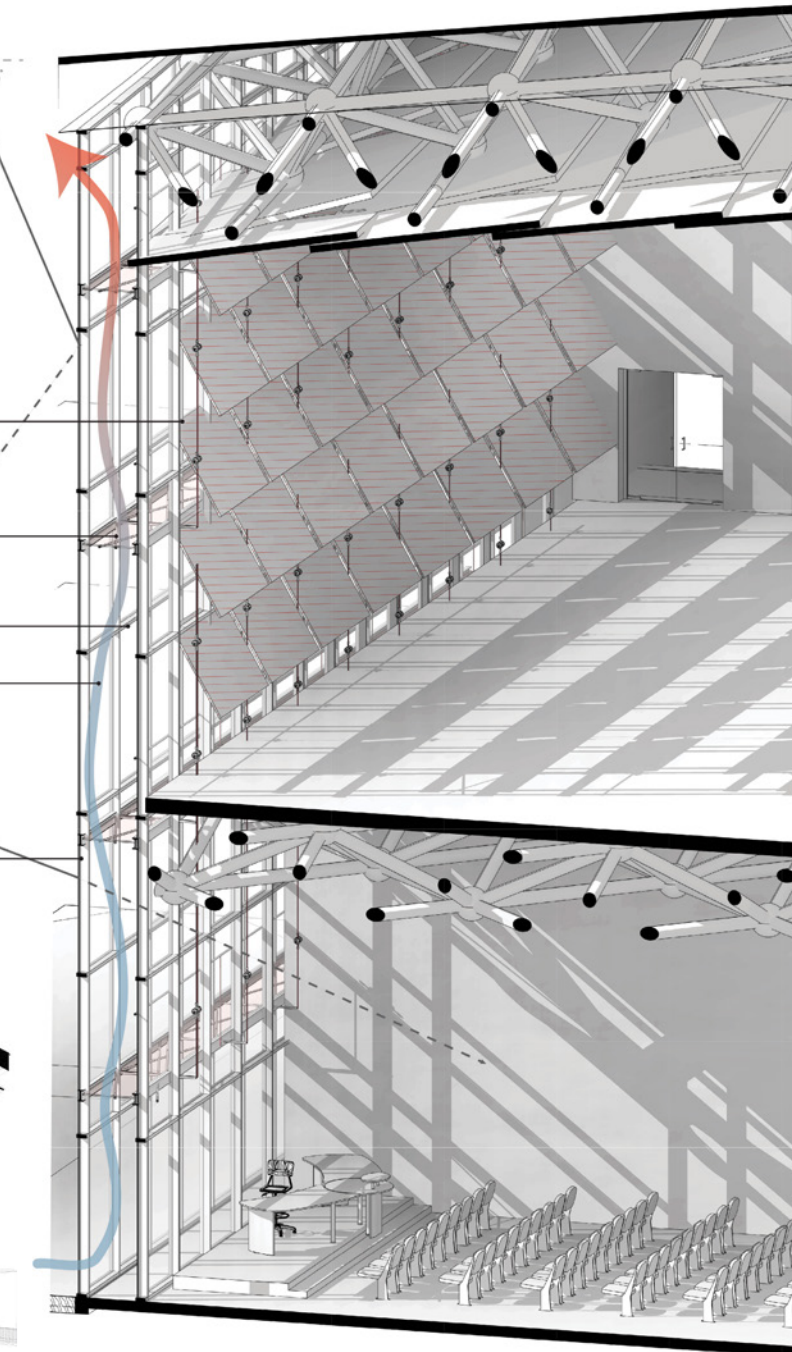
OPERABLE TIMBER
SUN SHADES MINI-
MIZE DIRECT SUN-
LIGHT FROM ENTER-
ING WORKSPACES

GLASS FLOORING
FOR MANAGEMENT
OF THE FACADE

INSULATED GLASS
UNIT

CAVITY WALL VEN-
TILATED WITH
TENANT EXHAUST
AIR

MONOLITHIC
GLASS THAT
THERMALLY AND
ENVIRONMENTAL-
LY PROTECTS THE
INNER FACADE
FROM THE SUN



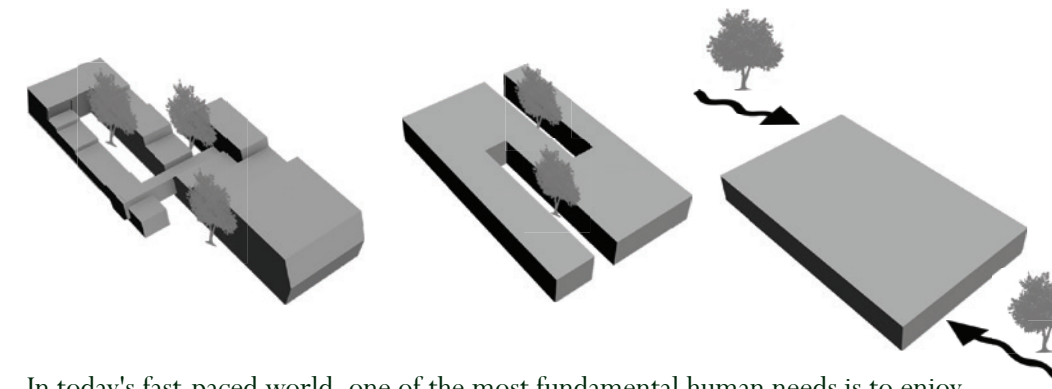
Library Design

Year : 2016

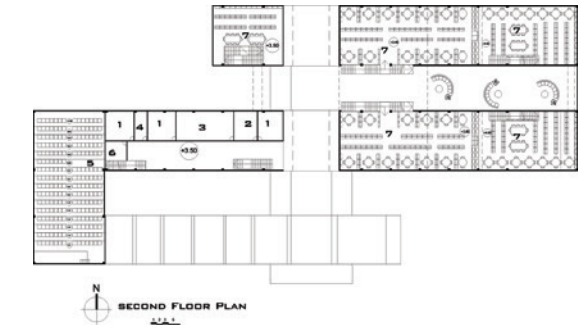
Type : Bachelor Individual Project

Location : Tabriz ,Iran

Professors :
Ali Farhadi

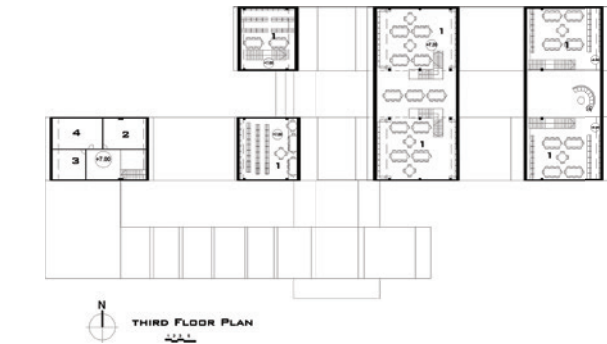


- 1 - OFFICE ROOM
- 2 - ACCOUNTANCY
- 3 - MEETING ROOM
- 4 - WORKPLACE
- 5 - AMPHITHEATER
- 6 - PANTRY
- 7 - STUDY

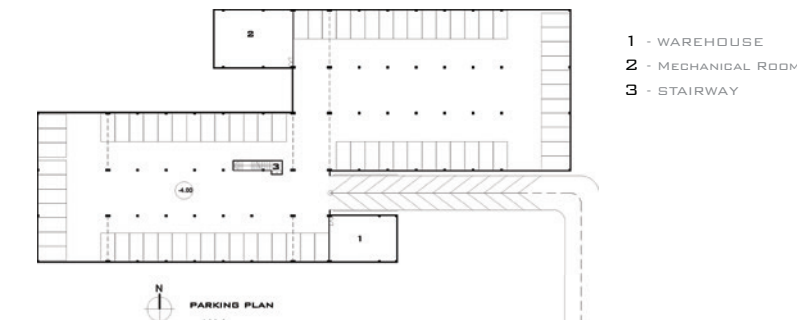


In today's fast-paced world, one of the most fundamental human needs is to enjoy nature and relax. A library can fulfill this need by providing a welcoming, inclusive, and relaxing environment. It is a place where people can visit to calm their minds. In my design, the most important elements were sunlight and a good view of the landscape.

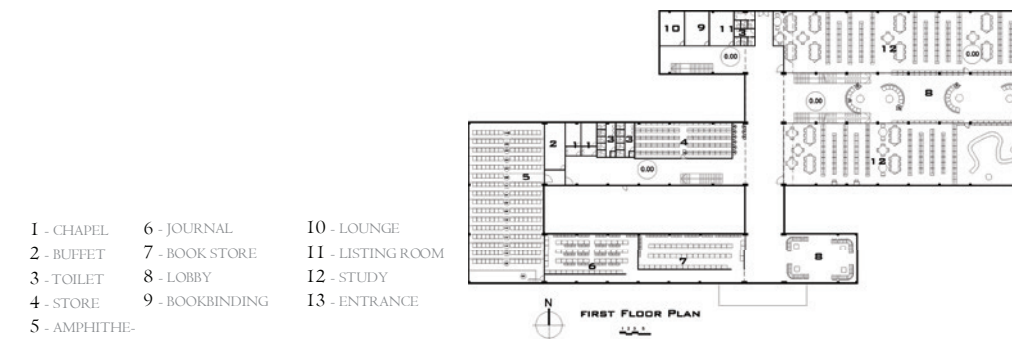
Nature plays a crucial role in separating the spaces, penetrating into them, and shaping the building. This integration creates more reading areas with excellent views of the surrounding landscape.



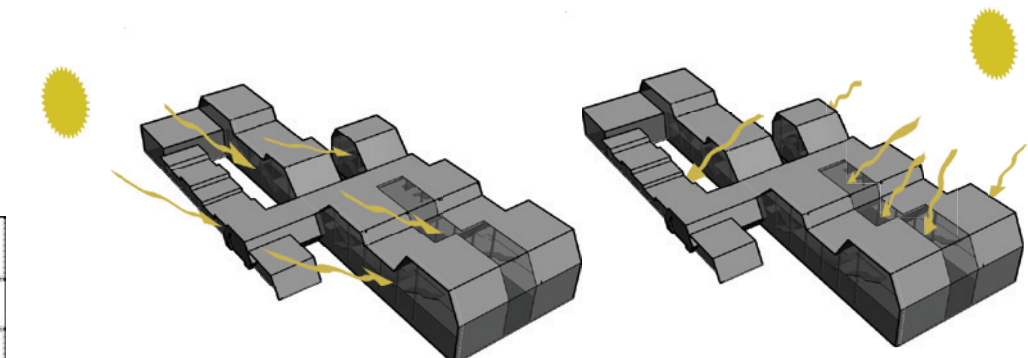
- 1 - STUDY
- 2 - ASSISTANT
- 3 - SECRETARY
- 4 - ADMINISTER



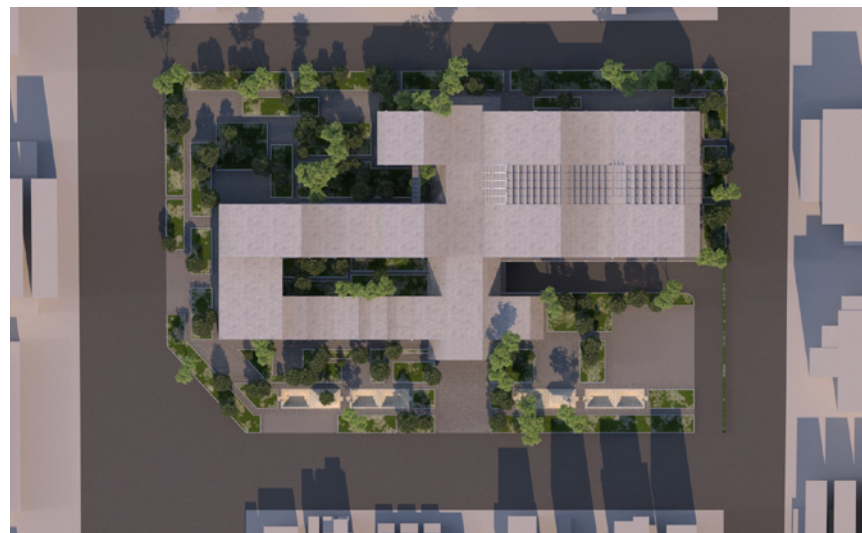
- 1 - WAREHOUSE
- 2 - MECHANICAL ROOM
- 3 - STAIRWAY



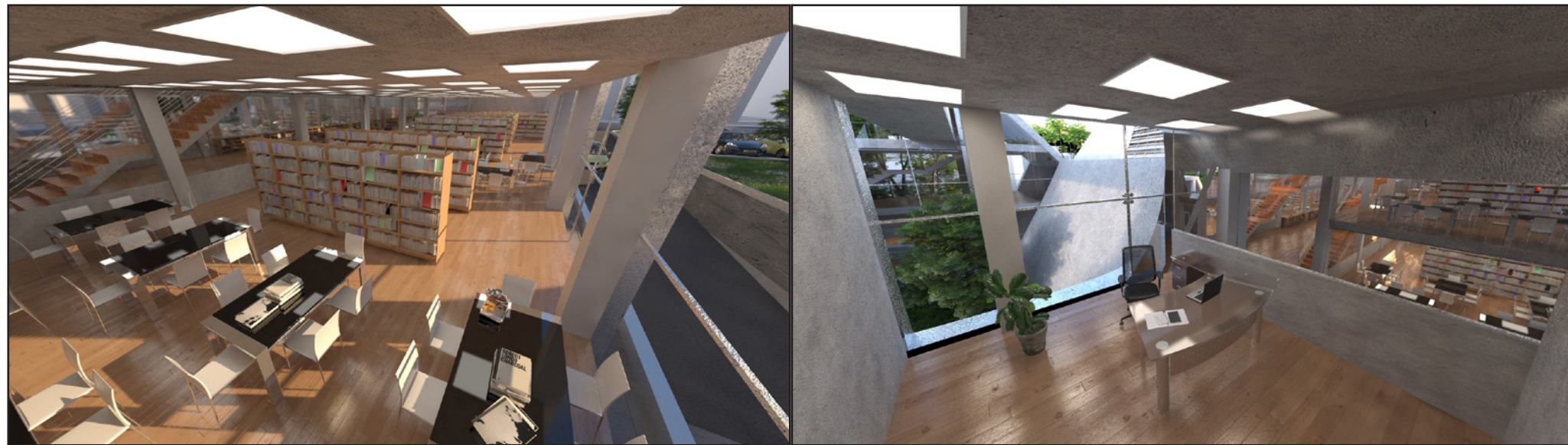
- 1 - CHAPEL
- 2 - BUFFET
- 3 - TOILET
- 4 - STORE
- 5 - AMPHITHE-
- 6 - JOURNAL
- 7 - BOOK STORE
- 8 - LOBBY
- 9 - BOOKBINDING
- 10 - LOUNGE
- 11 - LISTING ROOM
- 12 - STUDY
- 13 - ENTRANCE



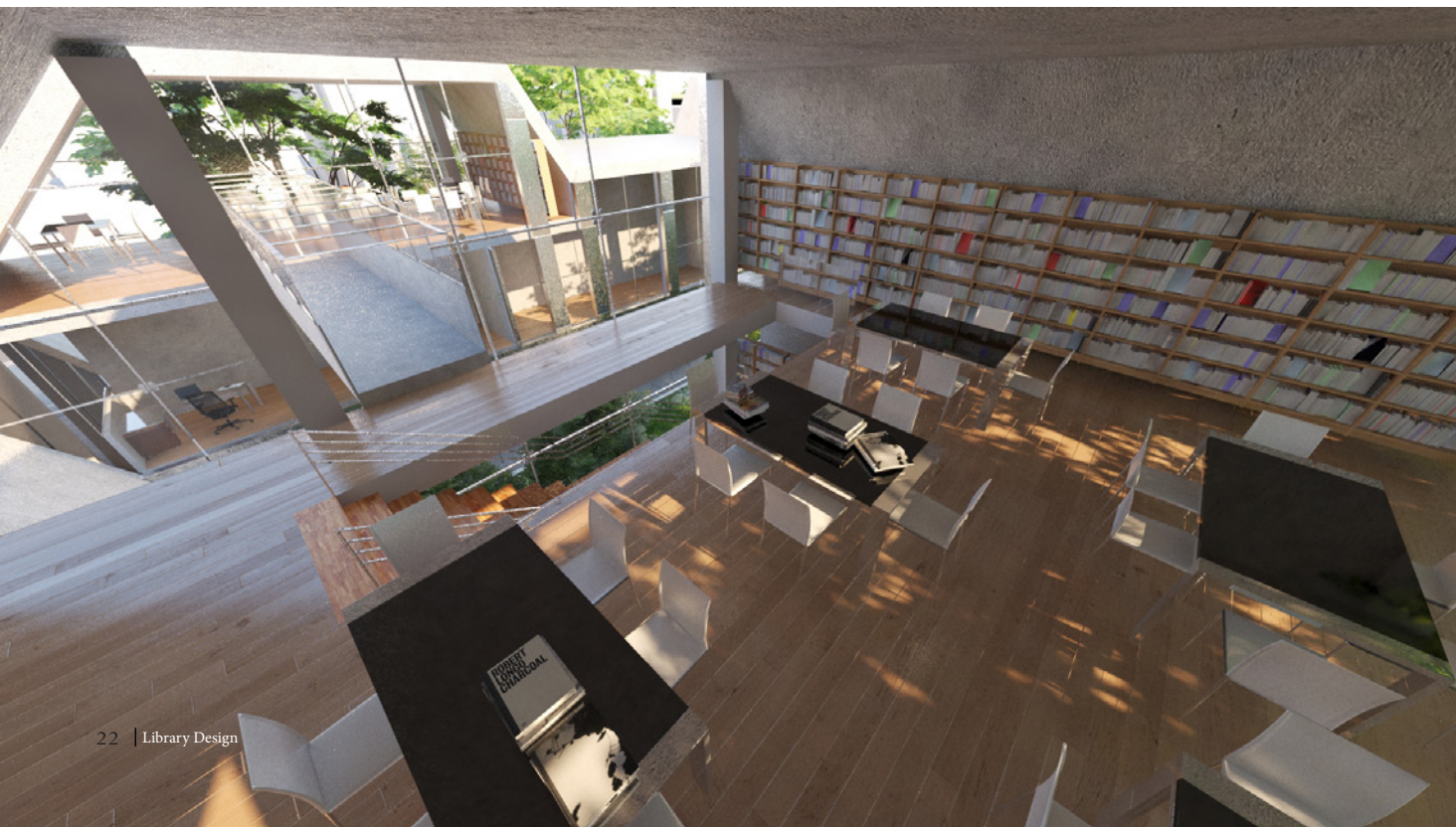
It is clear that northern and southern sunlight is the most favorable. I aimed to concentrate the majority of the building's glazing on the sun-facing facade to allow solar radiation directly into the space. To reduce the potential for overheating in the roof-glazing atrium, shading is provided by designing louvers.



Top Views



Connecting path Between Buildings





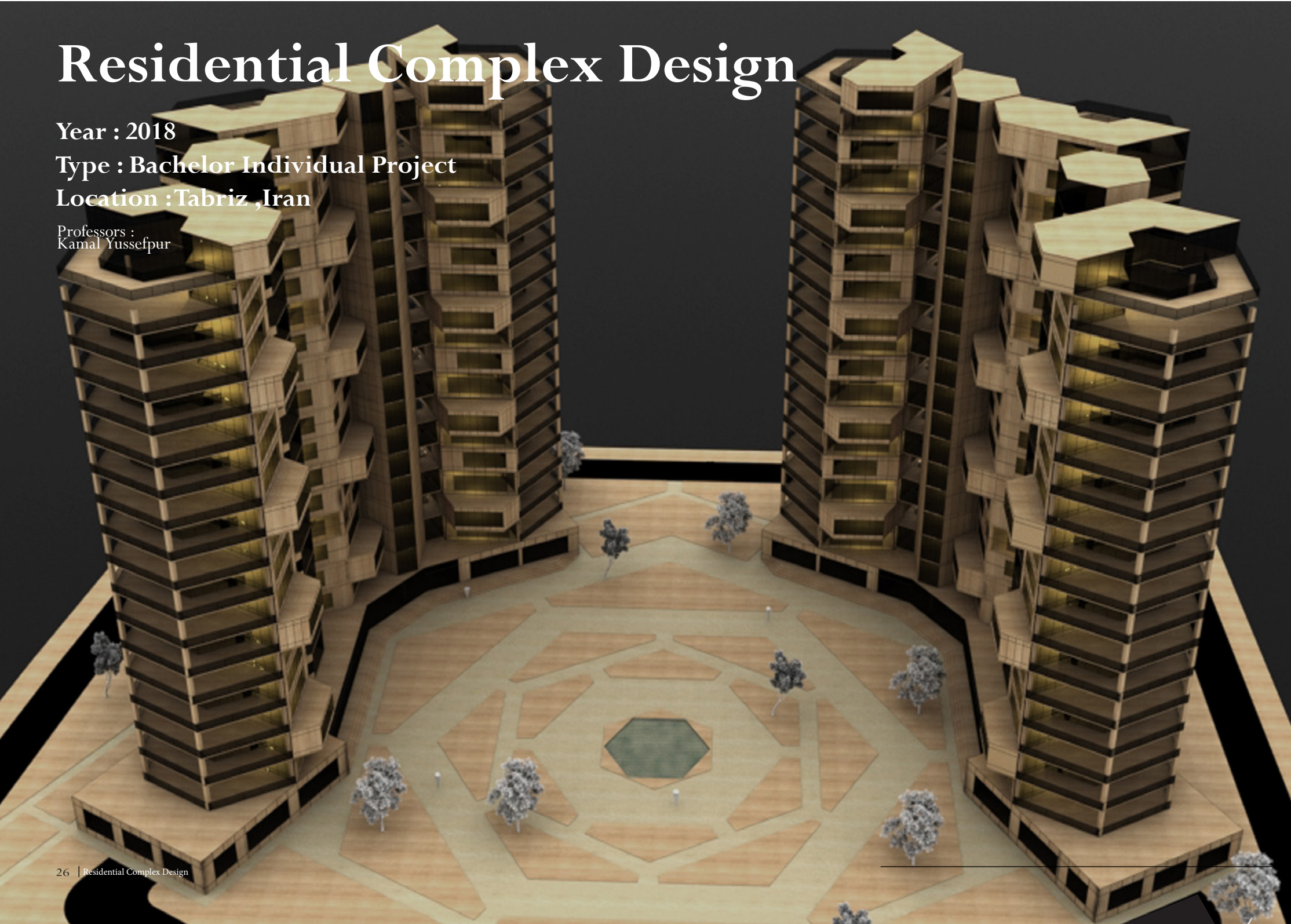
Residential Complex Design

Year : 2018

Type : Bachelor Individual Project

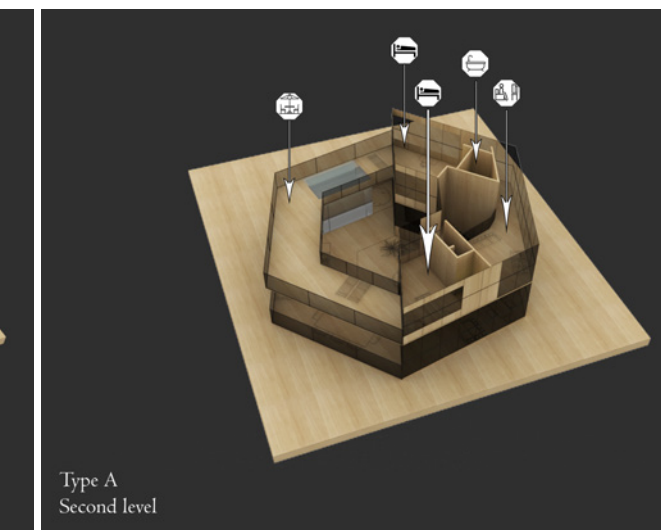
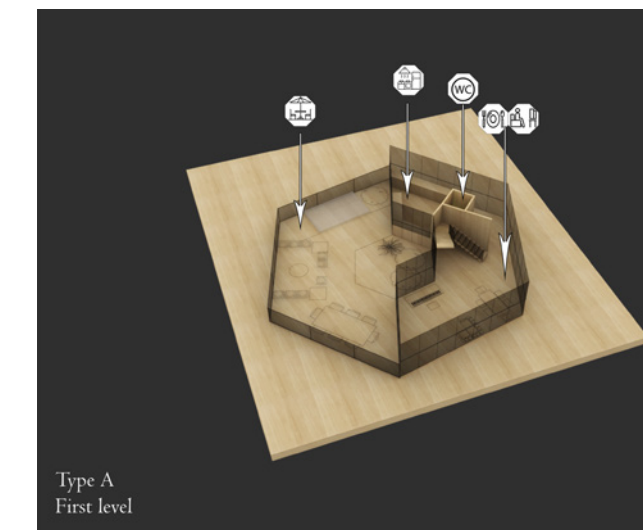
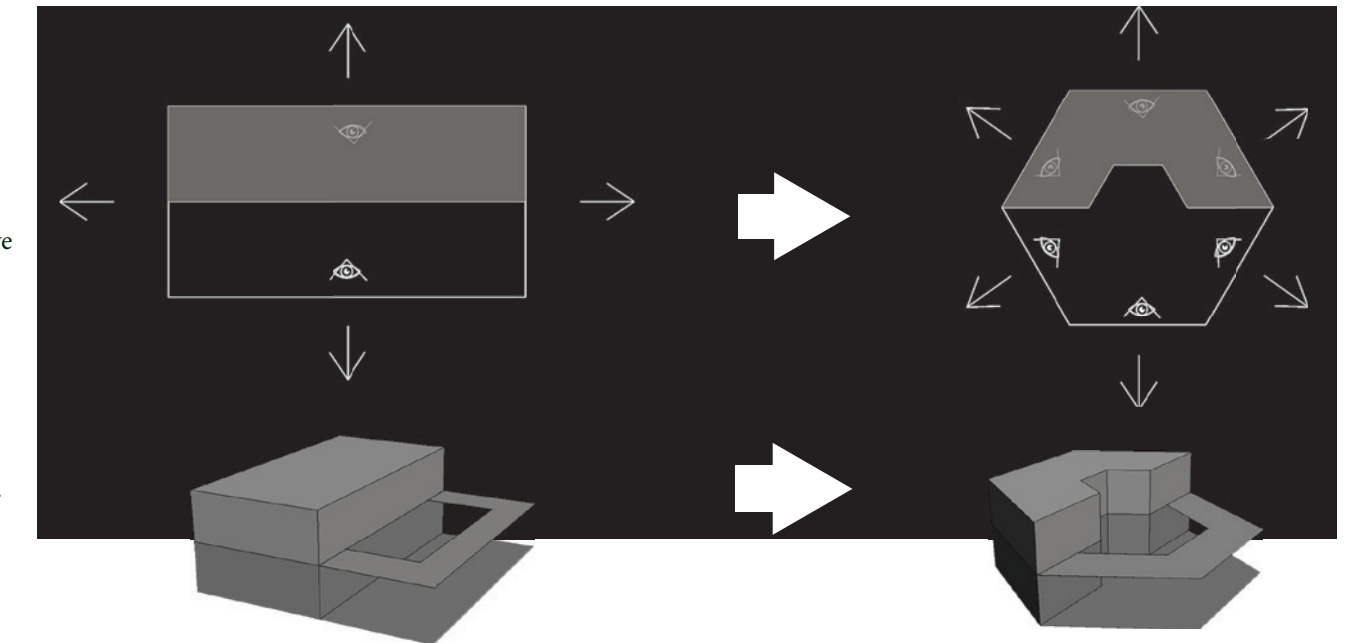
Location : Tabriz ,Iran

Professors :
Kamal Yussefpur

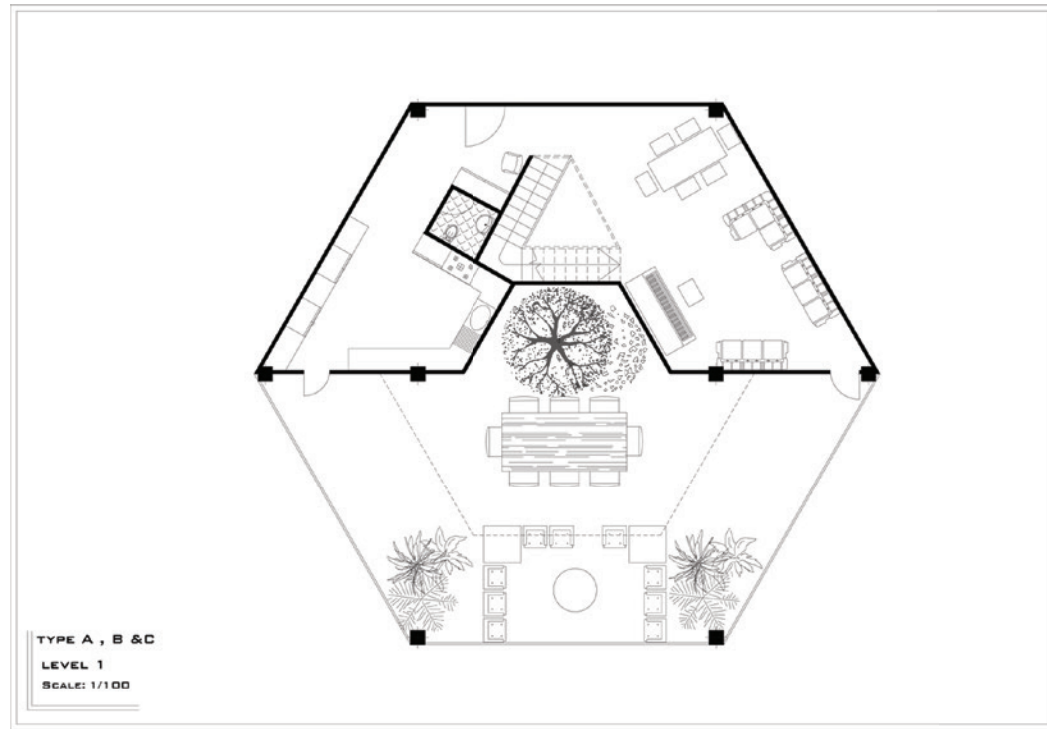
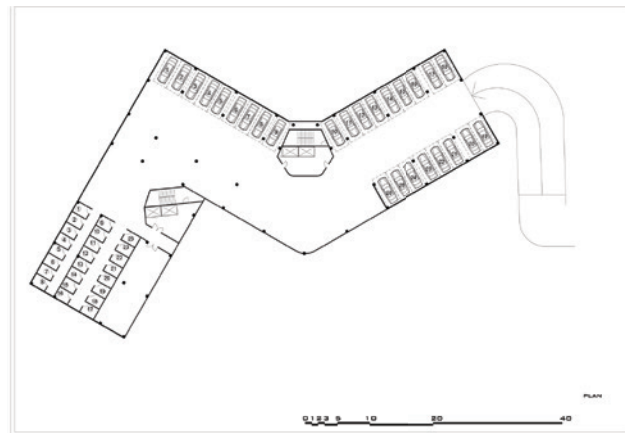


My design strategy centered on understanding and addressing the needs and desires of the residents and users. Residential complexes are often envisioned as monolithic structures with limited terraces, lacking a meaningful connection to nature and open skies. However, what if we reimagined this concept with a two-level block featuring expansive terraces? The diagram above demonstrates the transformation from a conventional residential unit to my innovative concept.

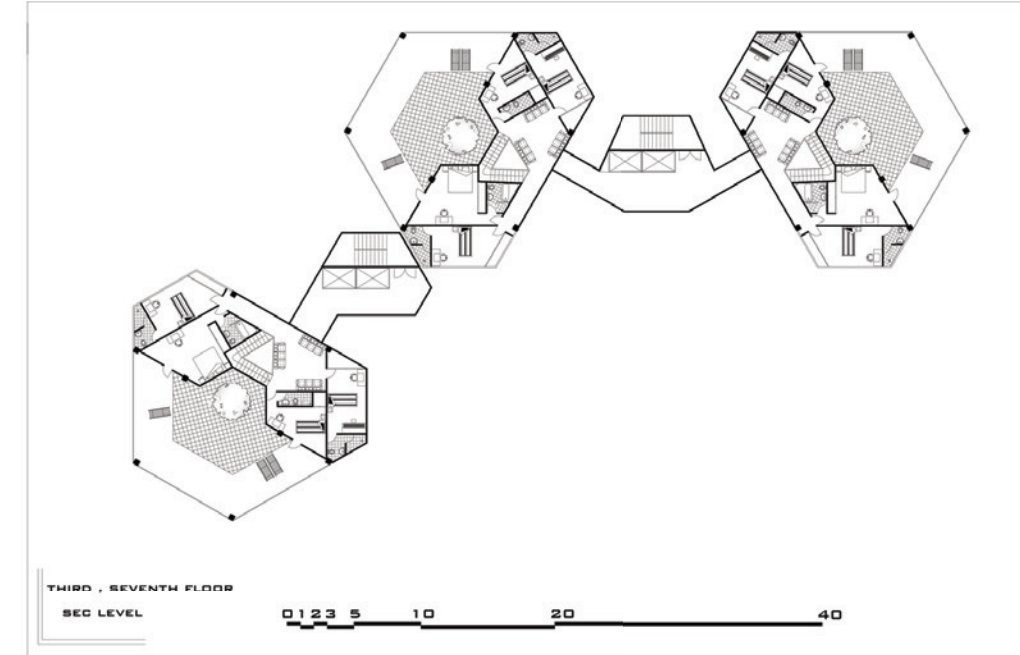
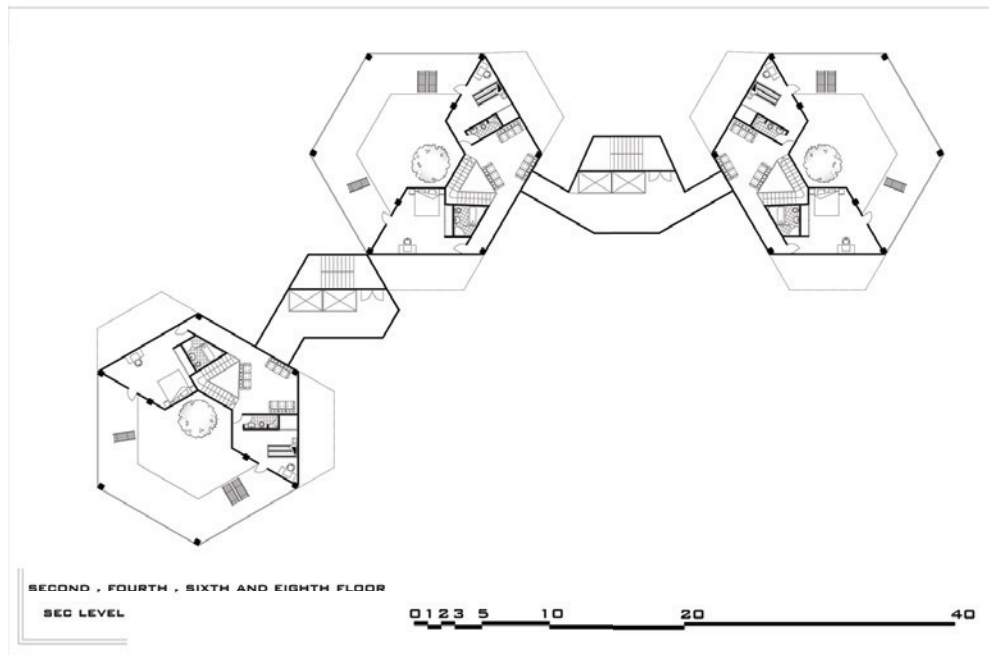
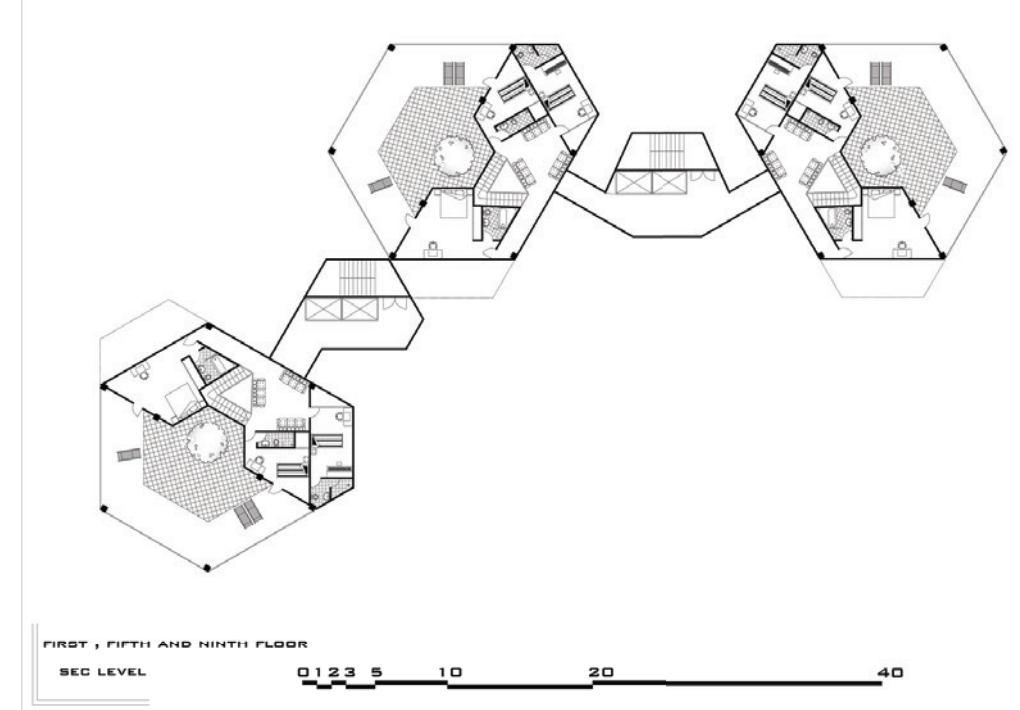
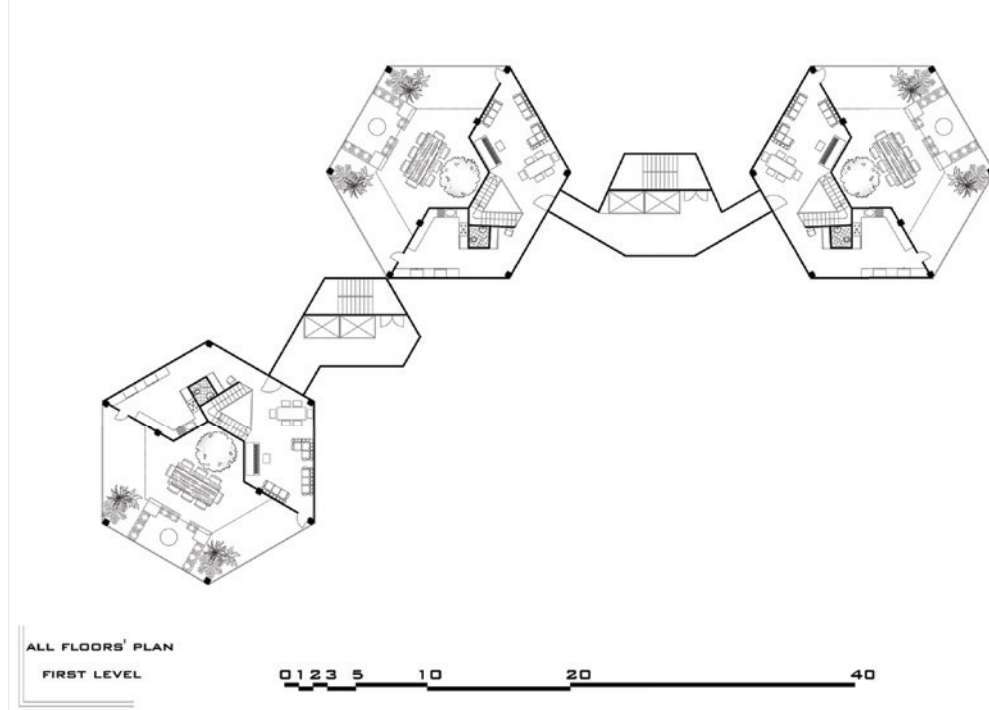
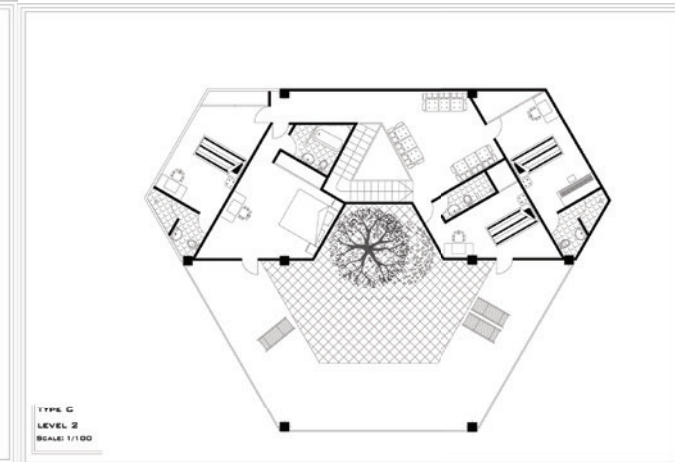
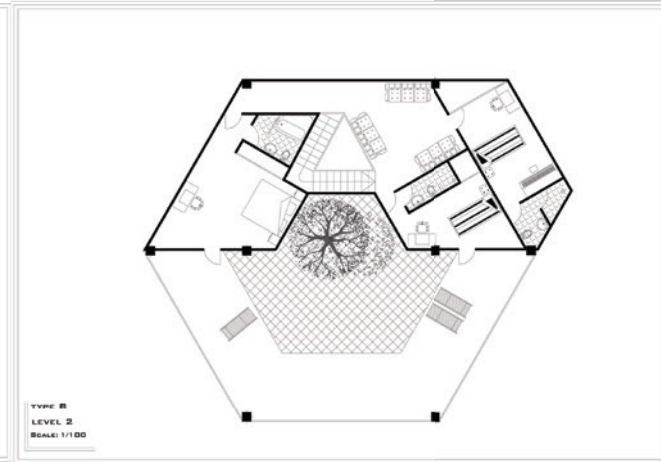
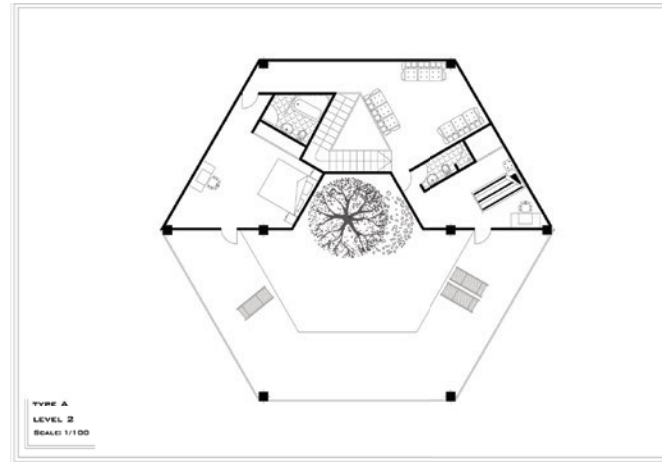
To achieve this, I opted for a hexagonal form, which offers more angles than traditional rectangular blocks. This geometry not only enhances the visual appeal but also maximizes views of the surrounding landscape, fostering a stronger connection with nature. The hexagonal design allows for panoramic vistas, creating an immersive experience for residents.

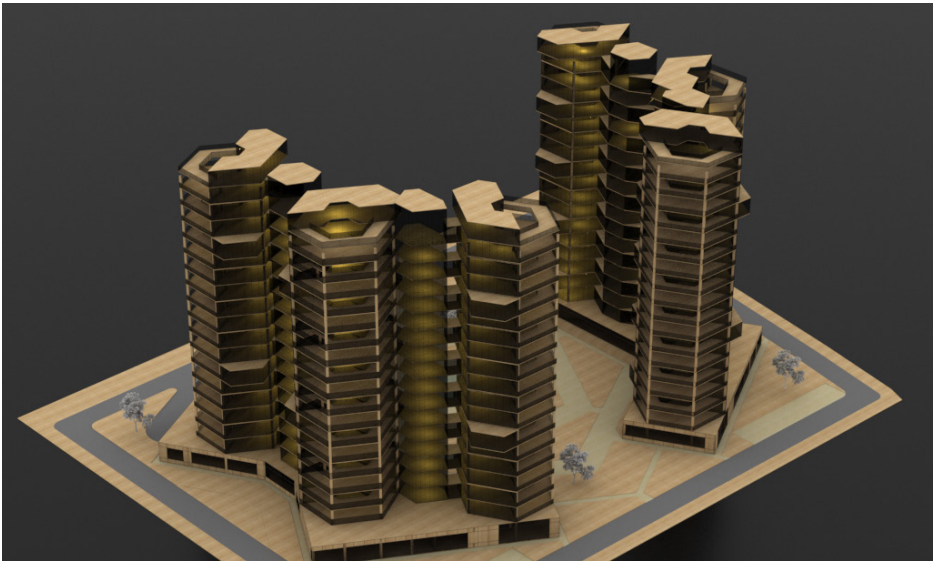
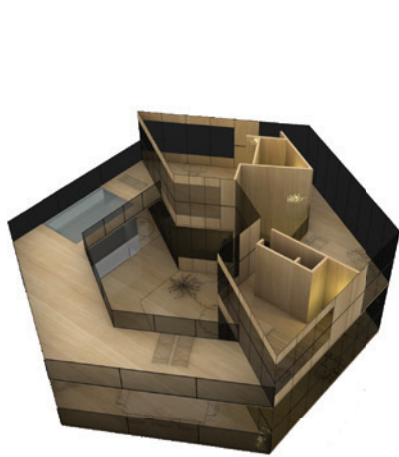
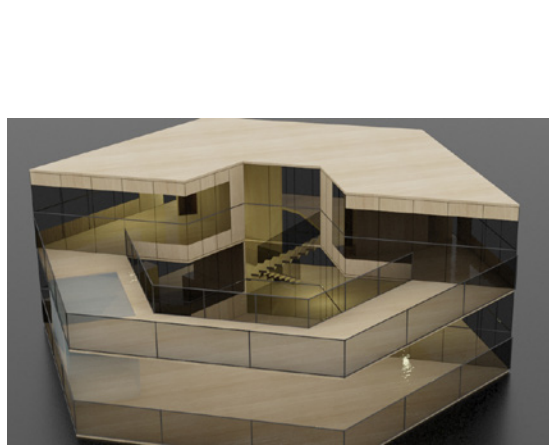
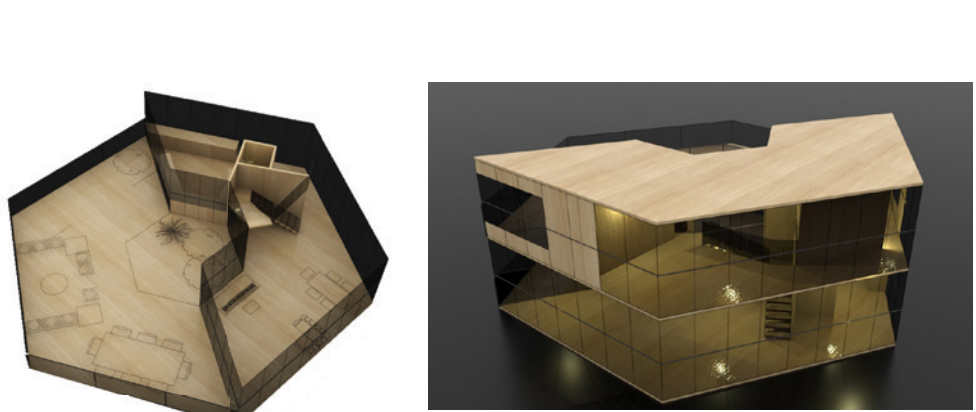
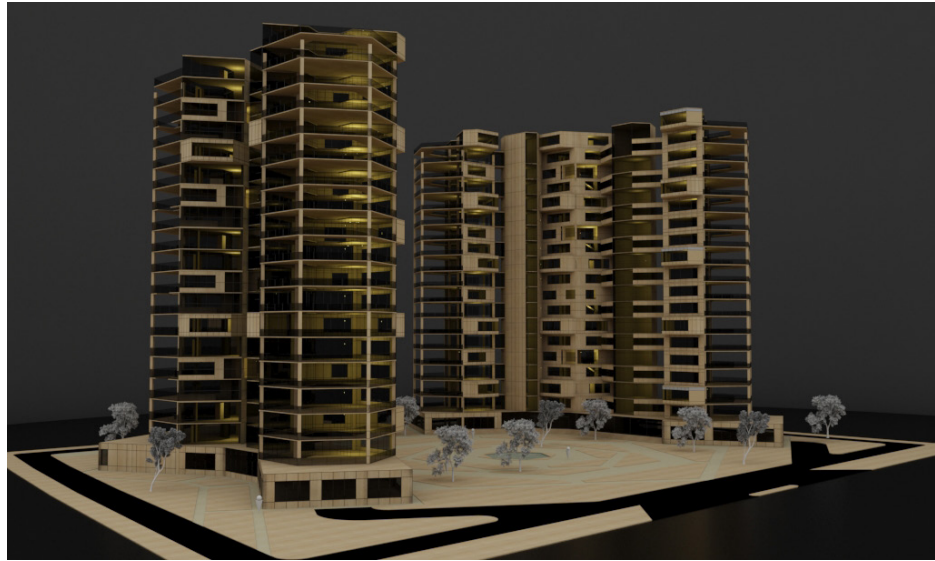
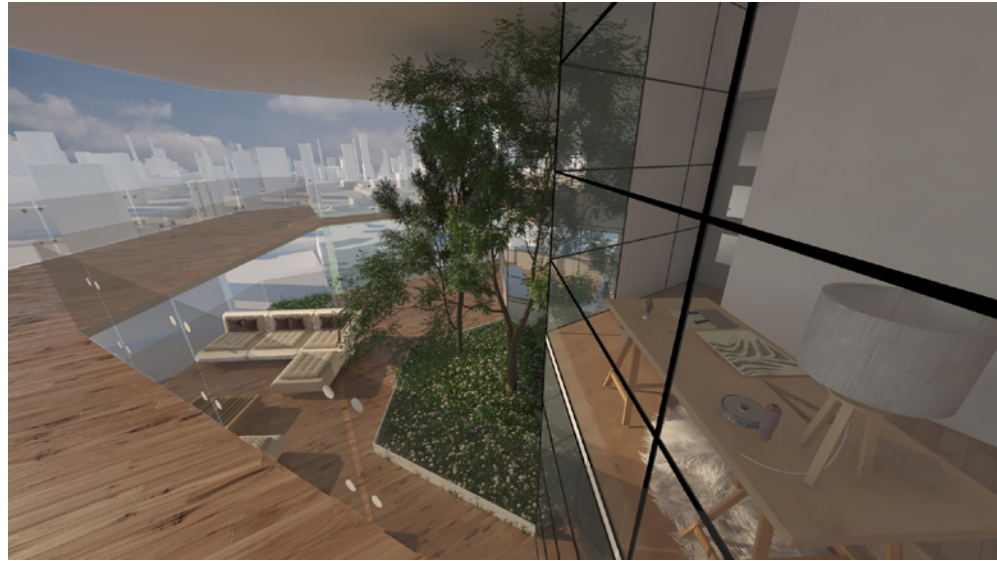


The residential complex, aptly named Hexagonal, is composed of blocks strategically positioned to provide unobstructed views of the lush landscape. This arrangement ensures that each unit benefits from natural light and scenic beauty, promoting a sense of well-being and harmony with the environment.



The design consists of three types of houses. The first level of all types is identical, but the second floor differs for each type. Type A is a single-room house, Type B is a double-room house, and Type C is a triple-room house.





Thank You

"I don't believe architecture has to speak too much. It should remain silent and let nature in the guise of sunlight and wind."

TADAO ANDO